

Chapter 1 : Floating Point Arithmetics

Fixed Point Representation

$$\begin{array}{lcl}
 + (10.1)_2 & \rightarrow & n=3 \\
 \begin{array}{c} \diagup \quad \diagdown \\ d_1 \quad d_2 \quad \frac{d_3}{K} \end{array} & & K=3 \\
 & & \beta = 2
 \end{array}
 \quad
 \begin{array}{lcl}
 - (123.12)_{10} & \rightarrow & n=5 \\
 \begin{array}{c} \diagup \quad \diagdown \quad \diagdown \quad \diagdown \\ d_1 \quad d_2 \quad d_3 \quad \frac{d_4}{K} \quad d_5 \end{array} & & K=4 \\
 & & \beta = 10
 \end{array}
 \quad
 \beta \text{ is the base}$$

$$X = \pm (d_1 d_2 \dots d_{K-1} \cdot d_K \dots d_n)_\beta$$

123.45

$(1001.11)_2$

left shift	right shift
12.345×10^1	1234.5×10^{-1}
0.12345×10^3	12345.0×10^{-2}

left shift	right shift
$(0.10011)_2 \times 2^4$	$(10011.1)_2 \times 2^{-1}$

Floating Point Representation (the computer way)

Convention - 1 : $F = \pm (0.d_1 d_2 \dots d_m)_\beta \times \beta^e$

(Standard/Generalized form)

fraction/mantissa
decimal part

e is exponent
 β is the base

CO1 : $d_1 = 1$ always to prevent multiple representation of a single number as it is a big chaos for computer. Also if every digit is 0 it is error.

As d_1 is fixed we can find out the combinations of d_2 and d_3

$$\therefore \text{Possible numbers generated} = 2^2 = 4$$

$$\therefore \text{Number of combinations} = 2^{\text{bit}}$$

Given $\beta = 2$, $m = 3$... Find out the Highest and the lowest
 $e_{\min} = -1$ or $[e = -1, 2]$ Possible Positive Number can this
 $e_{\max} = 2$ computer generate?

Highest Possible Positive Number $= +(0.111)_2 \times 2^2 = 7/2$

Lowest Possible Positive Number $= +(0.100)_2 \times 2^{-1} = 1/4$

Highest Possible Negative Number $= -(0.111)_2 \times 2^2$

Lowest Possible Negative Number $= -(0.100)_2 \times 2^{-1}$

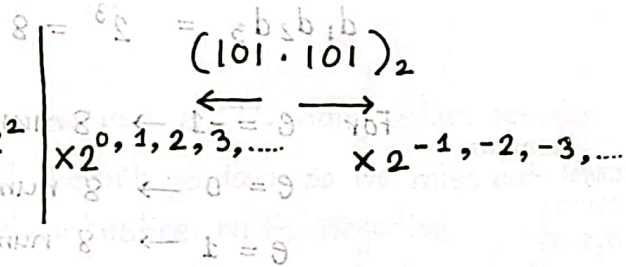
Highest Possible Number $= \pm (0.111)_2 \times 2^2$

Lowest Possible Number $= \pm (0.100)_2 \times 2^{-1}$

Here, $(0.111)_2 \times 2^2$

$= (1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}) \times 2^2$

$= 7/2$



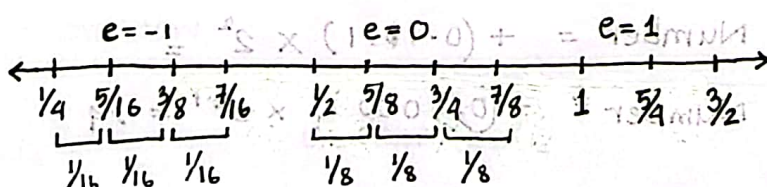
What are the next numbers in decimal and draw number line. (16 to)

$e = -1$ $\left[\begin{array}{l} (0.100)_2 \times 2^{-1} = 1/4 \\ (0.101)_2 \times 2^{-1} = 5/16 \\ (0.110)_2 \times 2^{-1} = 3/8 \\ (0.111)_2 \times 2^{-1} = 7/16 \end{array} \right.$

$e = 1$ $\left[\begin{array}{l} (0.100)_2 \times 2^1 = 1 \\ (0.101)_2 \times 2^1 = 5/4 \\ (0.110)_2 \times 2^1 = 3/2 \\ (0.111)_2 \times 2^1 = 7/4 \end{array} \right.$

$e = 0$ $\left[\begin{array}{l} (0.100)_2 \times 2^0 = 1/2 \\ (0.101)_2 \times 2^0 = 5/8 \\ (0.110)_2 \times 2^0 = 3/4 \\ (0.111)_2 \times 2^0 = 7/8 \end{array} \right.$

$e = 2$ $\left[\begin{array}{l} (0.100)_2 \times 2^2 = 2 \\ (0.101)_2 \times 2^2 = 5/2 \\ (0.110)_2 \times 2^2 = 3 \\ (0.111)_2 \times 2^2 = 7/2 \end{array} \right.$



the gaps between e are changing in an exponent power. equal spaced numbers as difference same

Convention - 2 : $F = \pm (1.d_1d_2d_3 \dots d_m)_\beta \times \beta^e$
(Normalized Form)

$\beta = 2, m = 3$

$$e_{\min} = -1$$

$$e_{\max} = 2$$

Highest Possible Positive Number = $+(1.000)_2 \times 2^{-1} = \frac{1}{2}$

Lowest Possible Positive Number = $+(1.111)_2 \times 2^2 = 7$

How many numbers can it generate? (64)

$$d_1d_2d_3 = 2^3 = 8$$

For $e = -1 \rightarrow 8$ numbers

$e = 0 \rightarrow 8$ numbers

$e = 1 \rightarrow 8$ numbers

$e = 2 \rightarrow 8$ numbers

$\therefore 32$ numbers (positive)

Again, 32 numbers (negative)

$\therefore$ Total 64 numbers.

Convention - 3 : $F = \pm (0.1d_1d_2d_3 \dots d_m)_\beta \times \beta^e$
(Denormalized Form)

$\beta = 2, m = 3$

$$e_{\min} = -1$$

$$e_{\max} = 2$$

1 bit extra in mantissa

Highest Possible Positive Number = $+(0.1111)_2 \times 2^2 = \frac{7}{4}$

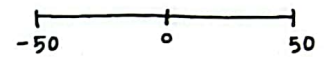
Lowest Possible Positive Number = $+(0.1000)_2 \times 2^{-1} = \frac{1}{4}$

How many numbers can it generate?

Same as convention 2.
64 numbers.

Calculate smallest possible negative value

Trick : Calculate highest value and put negative sign in front of it as 2



IEEE Standard (1985)

64 bits
 $\beta = 2$
 52 bits mantissa/fraction/Precision
 11 bits for exponent $\rightarrow$ can be $2^{11} = 2048$ exponent
 1 bit for sign

Initial Normalized form $\rightarrow F = \pm (1.d_1d_2 \dots d_{52})_2 \times 2^e$

$e_{min} = 0$
 $e_{max} = 2047$

Value of exponent

$e_{min} = 0$

$e_{max} = 2047$

When we use e^{2047} , many values remain unused.
 The value becomes 1 and it doesn't go down so we miss out lower values.
 So we divide and do injustice with negative values.
 called exponent biasing.

exponent biasing $\rightarrow$ We can set minimum and maximum value by this

$e_{min} = -1023$

$e_{max} = 1024$

So we use $\rightarrow e - 1023$

e^{-1023} is also wrong as e to the power gets

a minus value in e_{min}

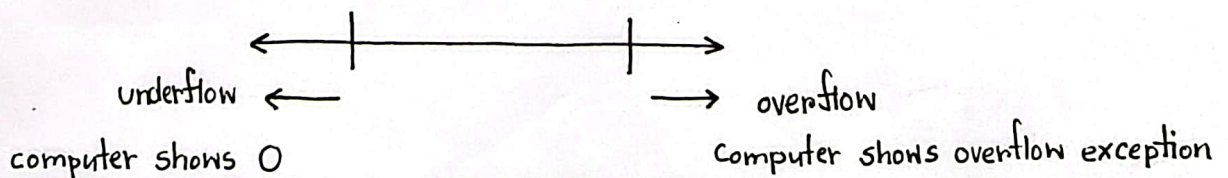
exponent biasing: $1024 \mid 0, 1023$

$\therefore$ Final Normalized form $\rightarrow F = \pm (1.d_1d_2 \dots d_{52})_2 \times 2^{e-1023}$

Now, we can use, $e_{min} = 0$

$e_{max} = 2047$

- Exponent biasing করার পর 3 আমাদের নিচের দিকের value আসলে ঠোঁট exponential underflow. উদয়ের দিকের value আসলে ঠোঁট exponential overflow.



Denormalized form :

$$(0.1d_1d_2\dots d_{52})_2 \times 2^{\underbrace{e-1023+1}_{2^{e-1022}}}$$

$\beta = 2$ $m = 3$ $e_{\min} = 0$

$e_{\max} = 2047$

largest possible positive number = $(0.\underbrace{1111\dots1}_{52 \text{ to } 1})_2 \times 2^{2047-1022}$

= $(0.111\dots1)_2 \times 2^{1025}$

lowest possible positive number = $(0.\underbrace{10000\dots0}_{52 \text{ to } 0})_2 \times 2^{0-1022}$

= $(0.1000\dots0)_2 \times 2^{-1022}$

For a computer $\rightarrow$

2^{1025}

2^{-1022}

largest possible number except infinity = $\pm (0.1111\dots1)_2 \times 2^{1024}$

lowest possible number except zero = $\pm (0.100\dots0)_2 \times 2^{-1021}$

e এর value change করা exponent biasing, infinity calculation

bit number 52 থেকে change করে দিলে largest, lowest



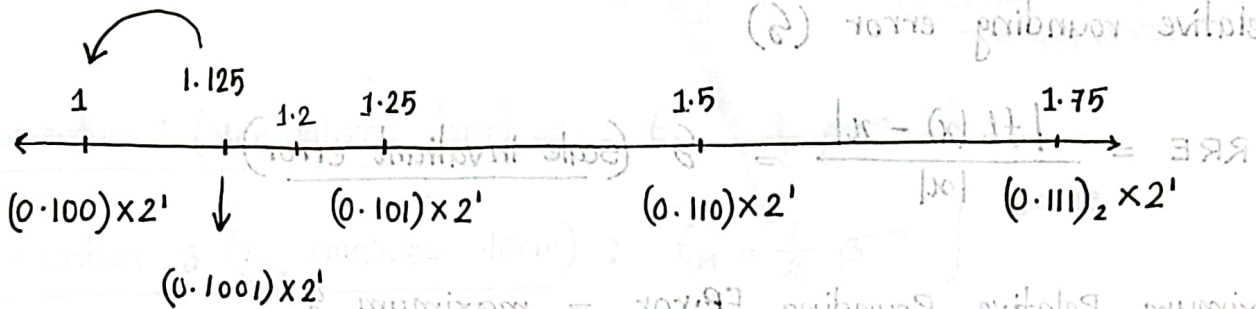
• Rounding

$\beta = 2$ $m=3$

$e_{min} = 1$

$e_{max} = 2$

convention 1 $\rightarrow$



input হিসেবে দিয়া হচ্ছে $\rightarrow x = 1.2 \rightarrow$ actual number

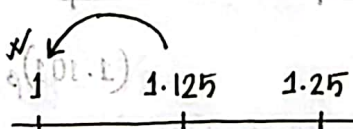
rounded (floating point representation) $\rightarrow fl(x) = 1.25$

1.1 যাবে 1 এ

1.2 যাবে 1.25 এ

1.125 যাবে $\downarrow$

★ if number is in the middle



round it to the nearest even number

	last digit
even -	0
odd -	1

• Number always serially আসবে as even-odd-even-odd কোনোদিনই even-even or odd-odd হবে না

- Rounding error

rounded value - exact value = rounding error



absolute value (+, -) doesn't matter

$$RE = |fl(x) - x| \quad \text{or} \quad fl(x) = x(1 + \delta)$$

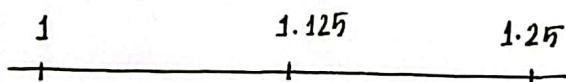
- Relative rounding error (δ)

$$RRE = \frac{|fl(x) - x|}{|x|} = \delta \quad (\text{scale invariant error})$$

- Maximum Relative Rounding Error = maximum δ
 $= \epsilon_m$ (machine epsilon)
 $=$ maximum possible invariant error

Convention 1

$$\delta \leq \epsilon_m$$



digit	test	$(0.100)_\beta \times \beta^e$
0	-	000
1	-	000

$$\text{Mid point distance} = \frac{1}{2} \times [(1.101)_\beta \times \beta^e - (0.100)_\beta \times \beta^e]$$

$$= \frac{1}{2} \times [(1.101)_\beta \times \beta^e - (0.100)_\beta \times \beta^e]$$

$$= \frac{1}{2} [2^{-3} \times \beta^e]$$

$$= \frac{1}{2} \beta^{-m} \beta^e \rightarrow \boxed{\text{max}} \text{ error } |fl(x) - x| \text{ at value}$$

$$|fl(x) - x| \leq \frac{1}{2} \beta^{-m} \beta^e$$

$$\begin{aligned} \text{minimum of } |x| &= (0.100)_2 \times \beta^e \\ &= 2^{-1} \times \beta^e \\ &= \beta^{-1} \times \beta^e \end{aligned}$$

minimum of x দিলে divide
দিলে maximum value লাগবে যা

$$\begin{aligned} \therefore \text{machine epsilon } (\epsilon_m) &= \frac{\frac{1}{2} \beta^{-m} \beta^e}{\beta^{-1} \beta^e} \rightarrow |fl(x) - x| \text{ (max)} \\ &\rightarrow |x| \text{ (min)} \\ &= \frac{1}{2} \beta^{1-m} \text{ (decimal এ আসবে)} \\ &\quad \text{কিছু করা লাগবে না} \end{aligned}$$

convention 2 (Normalized form) : $\epsilon_m = \frac{1}{2} \beta^{-m}$

convention 3 (Denormalized form) : $\epsilon_m = \frac{1}{2} \beta^{1-m}$ } same

• Floating Point Arithmetic with Rounding Error

$\beta = 2$ $m = 3$ (convention 1)

$e_{min} = -1$

$x = \frac{5}{8}$

$e_{max} = 2$

$y = \frac{7}{8}$

$\therefore x * y = ?$

$\therefore x = \frac{5}{8}$

$= (1.101)_2 \times 2^0$

$y = \frac{7}{8}$

$= (0.111)_2 \times 2^0$

$fl(x) = (0.101)_2 \times 2^0$

$fl(y) = (0.111)_2 \times 2^0$

$\therefore x * y = fl(x) * fl(y)$

$= [(0.101)_2 \times 2^0] \times [(0.111)_2 \times 2^0]$

$= (0.100011)_2 \times 2^0$

need to take till d_3 as we following convention 1, $m = 3$

Dec to Bin

ভাগশেষ আসলে 1

ভাগশেষ না আসলে 0

$0 \times 0 = 0$
$0 \times 1 = 0$
$1 \times 0 = 0$
$1 \times 1 = 1$

$0 + 0 = 0$
$0 + 1 = 1$
$1 + 0 = 1$
$1 + 1 = 10$

$0 - 0 = 0$
$0 - 1 = 1$
$1 - 0 = 1$
$1 - 1 = 0$

if $(m+1)^{th}$ digit = 0 $\rightarrow$ round it to previous number } middle ∇
 if $(m+1)^{th}$ digit = 1 $\rightarrow$ round it to next number } না হলে

(x) $(x+y) = (0.100)_2 \times 2^0 = (0.1)_2$

* power limit exceed কালে not possible to represent this number in convention 1

* স্মারক 1 না থাকলে আবার অন্য shifting করতে হবে.

• Significant figure $\rightarrow$ data precision/accuracy বাড়ানোর জন্য

4 significant figure (4sf)

left most non zero digit থেকে counting করতে হবে
 1. 55.996
 $4sf \rightarrow 56.00$
 $2sf \rightarrow 21$
 $1sf \rightarrow 20$

2.34921
 $4sf \rightarrow 2.350$
 2.34
 $4sf \rightarrow 2.340$

2.3496
 $4sf \rightarrow 2.350$ (rounded)
 0.00123
 $4sf \rightarrow 0.001230$ (non zero)

This is not decimal place তাই we don't count 0's.

• loss of significance

$$x^2 - 56x + 1 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x_1 = 28 + 3\sqrt{87} = 28 + \sqrt{(3\sqrt{87})^2} = 28 + 27.9821 = 55.9821$$

$$x_2 = 28 - 3\sqrt{87} = 28 - \sqrt{(3\sqrt{87})^2} = 28 - 27.9821 = 0.01786$$

45f :

Chapter 2 : Polynomial Interpolation

$$x_1 = 28 + \sqrt{783} = 28 + 27.98 = 55.98$$

$$x_2 = 28 - \sqrt{783} = 28 - 27.98 = 0.02000$$

very close দুটি number এর মধ্যে

minus (-) করতে গেলেই loss of significance পাওয়া যায়

How to prevent loss of significance :

$$\alpha = 55.98$$

$$\alpha\beta = \frac{c}{a}$$

$$= \frac{1}{1}$$

$$= \frac{1}{\alpha}$$

$$= \frac{1}{55.98}$$

$$= 0.01786$$

$$ax^2 + bx + c = 0$$

$$\alpha + \beta = -b/a$$

$$\alpha\beta = c/a$$

machine learning pipeline

$$\frac{1}{1!} - \frac{c}{1!} = 0 = 0.02$$

$$0 = 1! \quad 1 = 1! \quad 0 = 2! \quad 0 = 2! \quad \frac{1}{1!} - \frac{c}{1!} = 0 = 0.02$$

↓ terms ↑ polynomial ↑ approx

$$\frac{c}{1!} - c = (x) \text{ এর}$$

$$\frac{c}{1!} - c = (x) \text{ এর}$$

Chapter 2 : Polynomial Interpolation

$$\underbrace{-2x^3 + 4}_{\text{term}}$$

polynomial

$$2P.22 = 2P.FS + 8S = 28F + 8S = 10$$

$$a_0 = 0$$

$$00050.0 \quad a_1 = 8P.FS - 8S = 28F - 8S = 10$$

$$a_2 = 0$$

$$a_3 = -2$$

must maintain the sequence

general polynomial $\rightarrow P_n(x) = a_0x + a_1x^1 + a_2x^2 + a_3x^3 + \dots + a_nx^n$

$\downarrow$ degree (n+1) coefficients coefficient/constant

$\nwarrow$ variable $\nearrow$ degree

• Taylor Series

$$= \sum_{k=0}^n a_k x^k$$

$\swarrow$ natural basis $\searrow$ coefficients/constant

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3 + \dots$$

Taylor series of $\sin(x_0)$ where $x_0 = 0$

$$f(x) = \sin(x) \rightarrow f(0) = 0$$

$$f'(x) = \cos(x) \rightarrow f'(0) = 1$$

$$f''(x) = -\sin(x) \rightarrow f''(0) = 0$$

$$f'''(x) = -\cos(x) \rightarrow f'''(0) = -1$$

These functions are infinite but polynomials are finite

$$f(x) = 0 + 1x(x-0) + 0 - \frac{1x(x-0)^3}{3!} + \dots$$

$$f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

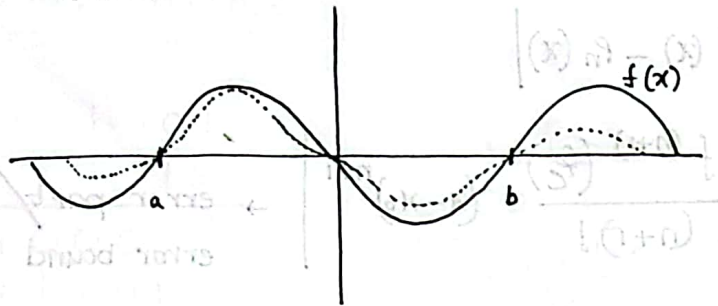
$$\begin{aligned} a_0 &= 0 & a_3 &= -\frac{1}{3!} \\ a_1 &= 1 & a_4 &= 0 \\ a_2 &= 0 & a_5 &= \frac{1}{5!} \end{aligned}$$

$$P_3(x) = x - \frac{x^3}{3!}$$

degree $\uparrow$ accuracy $\uparrow$ error $\downarrow$

$$P_5(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

• Theorem *



..... polynomial function
gap between and are the main error

Weierstrass Approximation Theorem or Truncation Error

$[a, b]$ maximum error ϵ

$f(x)$
 $P_n(x)$

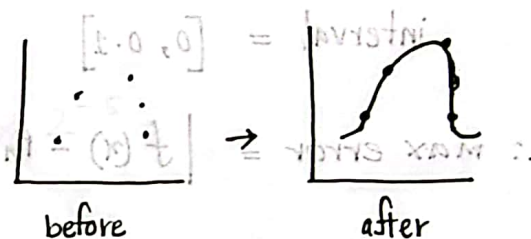
$$|f(x) - P_n(x)| \leq \epsilon$$

* theorem के लिए

explain with the graph and writings

• Polynomial Interpolation

There exists a $f(x)$ but we do not know how it looks like. But we know the values of some points such as $(x_1, y_1), (x_2, y_2), \dots$. After interpolation we will have a graph which will exactly go through the points.



• Taylor's theorem

Taylor series दिए error को define करा
or compact

generalized version of the Taylor series

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k$$

but putting n in place of ∞ will limit this.

This won't be infinite anymore. So some part is lost. lost part is called error

Taylor's error theorem:
$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k + \underbrace{\frac{f^{(n+1)}(\xi)}{(n+1)!} (x-x_0)^{n+1}}_{\text{error bound / error part / Lagrange form of remainder}}$$

error bound / error part / Lagrange form of remainder

$\xi(x_i)$ - ज्ञाते

↳ unknown variable

Layrange form of remainder = $|f(x) - P_n(x)|$

= $\left| \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-x_0)^{n+1} \right| \rightarrow \text{error part / error bound}$

$f(x) = \sin x$

$P_6(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$

if interval is given in ques, use that

if not $\rightarrow$ interval: $[x_0, x_1]$

$n = 6, x = 0.1$
 $x_0 = 0$ } ξ at min, max

Find out the maximum error

interval = $[0, 0.1]$

$\therefore \text{max error} = |f(x) - P_6(x)| = \left| \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-x_0)^{n+1} \right|$

= $\left| \frac{f^{(7)}(\xi)}{7!} (0.1-0)^7 \right|$

= $\left| \frac{-\sin(\xi)}{7!} (0.1)^7 \right|$

$\leq \left| \frac{\sin(0.1)}{7!} (0.1)^7 \right|$

try with lowest highest and middle point to get the highest value

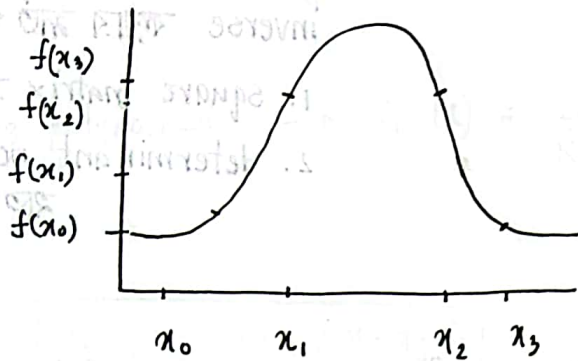
$\leq 1.98 \times 10^{-12}$ (put calc in radian)

points/nodes $\uparrow$ Accuracy $\uparrow$

$(n+1)$ point/nodes $\rightarrow$ polynomial degree n ?

Polynomial Interpolation methods

1. Vandermonde's Method / Matrix Method :



points are →

$$(x_0, f(x_0)), (x_1, f(x_1))$$

$$(x_2, f(x_2)), (x_3, f(x_3))$$

$$P_n(x) = a_0 x^0 + a_1 x^1 + a_2 x^2 + \dots + a_n x^n$$

$$f(x_0) = P_n(x_0) = a_0 x_0^0 + a_1 x_0^1 + a_2 x_0^2 + \dots + a_n x_0^n$$

$$f(x_1) = a_0 x_1^0 + a_1 x_1^1 + a_2 x_1^2 + \dots + a_n x_1^n$$

$$f(x_2) = a_0 x_2^0 + a_1 x_2^1 + a_2 x_2^2 + \dots + a_n x_2^n$$

⋮

$$f(x_n) = a_0 x_n^0 + a_1 x_n^1 + a_2 x_n^2 + \dots + a_n x_n^n$$

(n+1) nodes → (n+1) conditions

↳ degree n (always 1 less from node numbers)

Writing it in the matrix form →

$$x^1 = 1$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ 1 & x_2 & x_2^2 & \dots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix}$$

$$\underbrace{(n+1) \times (n+1)}_R \quad \underbrace{\quad}_C$$

$$\begin{bmatrix} f(x_0) \\ f(x_1) \\ f(x_2) \\ \vdots \\ f(x_n) \end{bmatrix}$$

This is the Vandermonde's matrix.

$$\therefore V \cdot A = F$$

$$\text{or } A = V^{-1} F$$

$$\# \quad x_0 = 0 \quad x_1 = \pi/2 \quad x_2 = \pi$$

$$f(x_0) = 1 \quad f(x_1) = 0 \quad f(x_2) = -1$$

3 nodes $\rightarrow P_2(x)$

$n = 2$ (degree)

$$P_2(x) = a_0 x^0 + a_1 x^1 + a_2 x^2$$

dimension (3×3)

$$\begin{bmatrix} 1 & x_0 & x_0^n \\ 1 & x_1 & x_1^n \\ 1 & x_2 & x_2^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} f(x_0) \\ f(x_1) \\ f(x_2) \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0^2 \\ 1 & \pi/2 & (\pi/2)^2 \\ 1 & \pi & \pi^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} f(x_0) \\ f(x_1) \\ f(x_2) \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & \pi/2 & \pi^2/4 \\ 1 & \pi & \pi^2 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2/\pi \\ 0 \end{bmatrix}$$

$$P_2(x) = 1x^0 - \frac{2}{\pi}x + 0 \cdot x^2$$

$$= 1 - \frac{2x}{\pi}$$

$$P_2(10) = 1 - \frac{2 \times 10}{\pi}$$

$$P_2(10) = 1 - \frac{20}{\pi}$$

inverse করার মত

1. square matrix হতে হবে

2. determinant non-zero হতে হবে

disadvantage

inverting a matrix is computationally very expensive

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

2. Lagrange Polynomial :

general/standard polynomial $\rightarrow P_n(x) = \sum_{k=0}^n a_k x^k = \prod_{i=0}^n \rightarrow$ product sign

lagrange polynomial $\rightarrow P_n(x) = \sum_{k=0}^n \underbrace{f(x_k)}_{\text{coefficient (known)}} \underbrace{l_k(x)}_{\text{lagrange basis (unknown)}}$

$$l_k(x) = \prod_{\substack{j=0 \\ j \neq k}}^n \left(\frac{x - x_j}{x_k - x_j} \right)$$

$$= \frac{(x - x_0)(x - x_1)(x - x_2) \dots (x - x_n)}{(x_k - x_0)(x_k - x_1)(x_k - x_2) \dots (x_k - x_n)}$$

disadvantage

• New nodes can't be added

$$\begin{aligned} x_0 &= -\pi/4 & x_1 &= 0 & x_2 &= \pi/4 \\ f(x_0) &= \frac{1}{\sqrt{2}} & f(x_1) &= 1 & f(x_2) &= \frac{1}{\sqrt{2}} \end{aligned}$$

• If added, then need to do whole $l_k(x)$ calculation newly again.

$$P_2(x) = f(x_0) l_0(x) + f(x_1) l_1(x) + f(x_2) l_2(x)$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{8}{\pi^2} x(x - \pi/4) + 1 \cdot \left(-\frac{16}{\pi^2}\right)(x + \pi/4)(x - \pi/4) + \frac{1}{\sqrt{2}} \cdot \frac{8}{\pi^2}$$

$$= \frac{16}{\pi^2} \left(\frac{1}{\sqrt{2}} - 1 \right) x^2 + 1$$

advantage

no need to inverse so saves money/cost

$$l_k(x) = \prod_{\substack{j=0 \\ j \neq k}}^n \left(\frac{x - x_j}{x_k - x_j} \right)$$

$$l_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 0)(x - \pi/4)}{(-\pi/4 - 0)(-\pi/4 - \pi/4)} = \frac{8}{\pi^2} x(x - \pi/4)$$

$$l_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x + \pi/4)(x - \pi/4)}{(0 + \pi/4)(0 - \pi/4)} = -\frac{16}{\pi^2} (x + \pi/4)(x - \pi/4)$$

$$l_2(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} = \frac{(x + \pi/4)(x - 0)}{(\pi/4 + \pi/4)(\pi/4 - 0)} = \frac{8}{\pi^2} x(x + \pi/4)$$

3. Newton's Polynomial

$$P_n(x) = \sum_{k=0}^n a_k n_k(x) = (x) \leftarrow \text{Newton's basis}$$

$\underbrace{\quad}_{\text{coefficient}} \quad \underbrace{\quad}_{\text{Newton's basis}}$

$$= a_0 n_0(x) + a_1 n_1(x) + a_2 n_2(x) + \dots + a_n n_n(x)$$

$$= f[x_0] \times 1 + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \dots$$

This is also called divided difference form.

$$\frac{\text{last node} - \text{first node}}{\text{last node} - \text{first node}} \rightarrow \frac{f[x_1] - f[x_0]}{x_1 - x_0} \quad \text{example}$$

$$n_0(x) = 1$$

$$n_1(x) = (x - x_0)$$

$$n_2(x) = (x - x_0)(x - x_1)$$

$$n_3(x) = (x - x_0)(x - x_1)(x - x_2)$$

⋮

$$a_0 = f[x_0]$$

$$a_1 = f[x_0, x_1]$$

$$a_2 = f[x_0, x_1, x_2]$$

$$a_3 = f[x_0, x_1, x_2, x_3]$$

$$x_0 = 1$$

$$x_1 = 0$$

$$x_2 = 1$$

$$x_3 = 2$$

$$f(x_0) = 5$$

$$f(x_1) = 1$$

$$f(x_2) = 1$$

$$f(x_3) = 11$$

$$P_3(x) = a_0 n_0(x) + a_1 n_1(x) + a_2 n_2(x) + a_3 n_3(x)$$

$$f[x_0] = 5$$

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = \frac{1 - 5}{0 - 1} = 4$$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = \frac{0 - 4}{1 - 1} = 2$$

$$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1} = \frac{5 - 0}{2 - 0} = 2.5$$

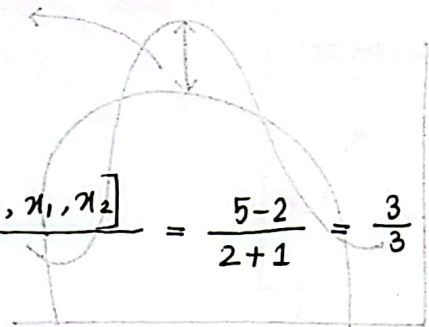
$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0} = \frac{2.5 - 2}{2 - 1} = 0.5$$

$$x_2 = 1 \quad f(x_2) = 1$$

$$f[x_2, x_3] = \frac{f[x_3] - f[x_2]}{x_3 - x_2} = \frac{11 - 1}{2 - 1} = 10$$

$$x_3 = 2 \quad f(x_3) = 11$$

$$* f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0} = \frac{5 - 2}{2 + 1} = \frac{3}{3} = 1$$



$$P_3(x) = a_0 n_0(x) + a_1 n_1(x) + a_2 n_2(x) + a_3 n_3(x)$$

$$= f[x_0] \times 1 + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + f[x_0, x_1, x_2, x_3](x - x_0)(x - x_1)(x - x_2)$$

$$= (5 \times 1) + (-4)(x + 1) + (2)(x + 1)(x - 0) + (1)(x + 1)(x - 0)(x - 1)$$

$$= 5 - 4(x + 1) + 2x(x + 1) + x(x + 1)(x - 1)$$

$$P_3(10) =$$

- This method is useful because it can easily add new node, saved previous process in storage.
- benefit of newton's method → reusability.

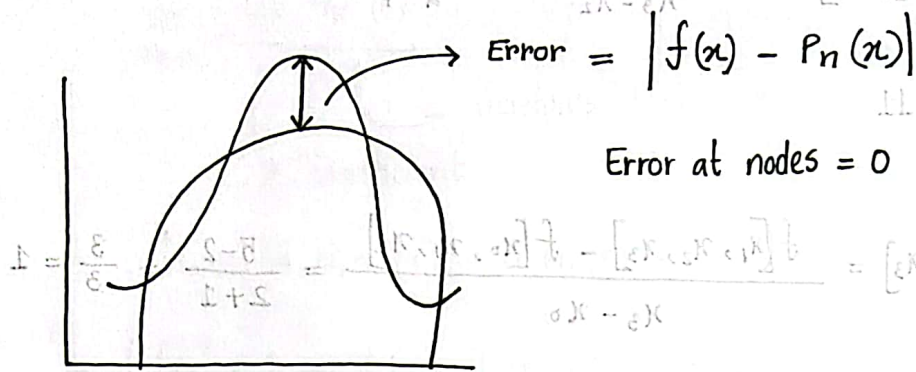
lagrange method এ পুনরাবৃত্তি আবার করতে হয় if we add new node.

Quiz 2 → 18th february

chapter 2

till chabyshev nodes

• Interpolation Error (For Multiple Nodes)



cauchy's theorem

$$x_0, x_1, x_2, \dots, x_n \in [a, b]$$

$$f(x) \rightarrow P_n(x)$$

$$|f(x) - P_n(x)| = \left| \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-x_0)(x-x_1)(x-x_2)\dots(x-x_n) \right|$$

$$\# f(x) = \cos x$$

$$x_0 = -\pi/4 \quad x_1 = 0 \quad x_2 = \pi/4 \quad [-1, 1]$$

$$f(x_0) = 1/\sqrt{2} \quad f(x_1) = 1 \quad f(x_2) = 1/\sqrt{2}$$

nodes দিয়া আছে $\rightarrow 3$

$$\therefore \text{degree} = 3-1 = 2 = n$$

Find max error?

$$|f(x) - P_2(x)| = \left| \frac{f^{(3)}(\xi)}{3!} (x+\pi/4)(x-0)(x-\pi/4) \right|$$

$$= \left| \frac{\sin(\xi)}{6} \cdot x(x+\pi/4)(x-\pi/4) \right|$$

ξ এর range $[-1, 1]$

$$\leq \frac{\sin(1)}{6} \times 0.383 \rightarrow \text{max function value}$$

max interval value

$$W(x) = x(x + \pi/4)(x - \pi/4)$$

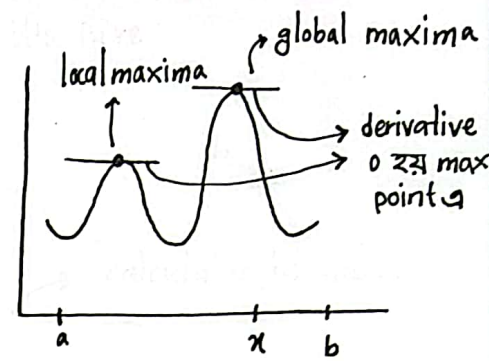
$$= x(x^2 - \pi^2/16)$$

$$= x^3 - \pi^2/16 x$$

$$W'(x) = 3x^2 - \pi^2/16 = 0$$

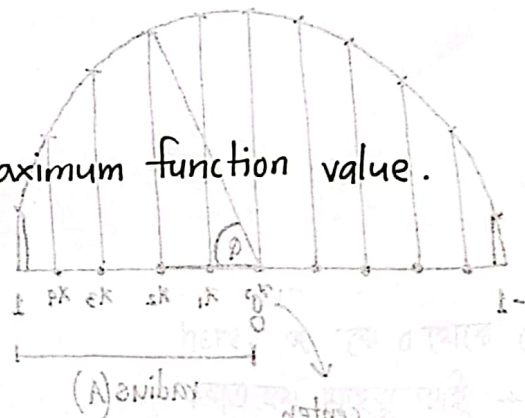
$$\Rightarrow x = \pm \pi/4\sqrt{3}$$

to find max value we need to check the two interval values and the two values of x .
Total 4 values



x	$W(x)$
$-\pi/4\sqrt{3}$	$+0.186$
$\pi/4\sqrt{3}$	-0.186
-1	-0.383
$+1$	$+0.383$

this is our maximum function value.



Chabyshev Nodes

Generally $\rightarrow$ Nodes $\uparrow$ Error $\downarrow$

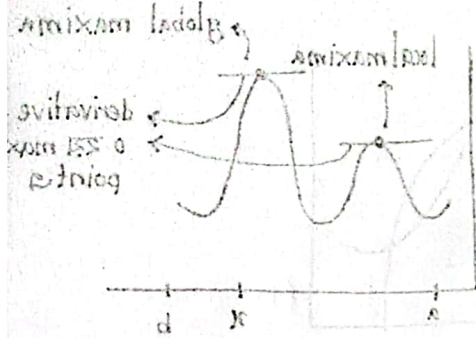
But exception $\rightarrow$ Nodes $\uparrow$ Error $\uparrow$ — called Runge Phenomenon

function is called Runge Function

Example : $f(x) = \frac{1}{1+25x^2}$ $[-1, 1]$

So the solution of the problem $\rightarrow$ chabyshev

অব node এ হয়না
equally spaced nodes
অনেক বেশি দরিয়ান nodes আগে



$$W(x) = (x^2 - A^2)(x^2 + A^2) = 0$$

$$(x^2 - A^2)(x^2 + A^2) = 0$$

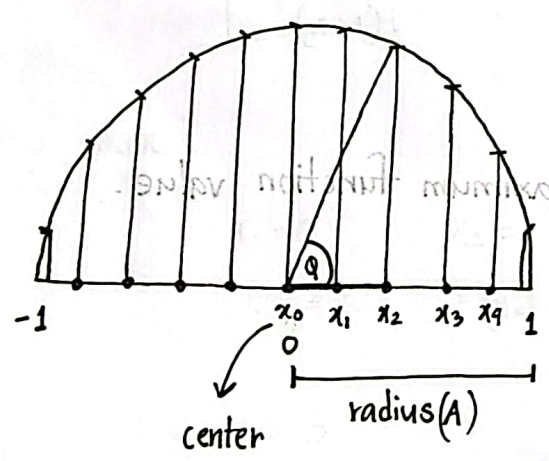
$$x^2 - A^2 = 0$$

$$x^2 = A^2$$

$$x = \pm A$$

out of the two values we need to check the two interval values and the two values of x .
 Solution to this issue →

1. Do not take equally spaced nodes - but how?
2. Take more nodes at the corner



can't take straight lines
 কারণ দূরত্ব জায়ে equally spaced হয়ে যাবে

so we take circle and draw nodes here → not equally spaced edge এর দিকে density বেশি

For the value of x

$$\cos(\phi) = \frac{x}{A}$$

$$\Rightarrow x = A \cos(\phi)$$

$$\therefore x_j = A \cos\left(\frac{(2j+1)\pi}{2(n+1)}\right)$$

But exception → Nodes $j = 0, 1, 2, 3, \dots, n$

$$\phi_j = \frac{(2j+1)\pi}{2(n+1)}$$

$f(x) = \frac{1}{1+25x^2}$ and interval $[-1, 1]$

Make 4 chebyshev nodes.

আম্রাল function এর
কোনো কাজ নাই হুদাই
it's here

degree, $n = 4 - 1 = 3$

radius, $A = 1 \rightarrow$ as the interval is $[-1, 1]$

calculator in radian

$$x_0 = 1 \cos \frac{(2 \times 0 + 1) \pi}{2(3+1)} = \cos \frac{\pi}{8} = 0.92$$

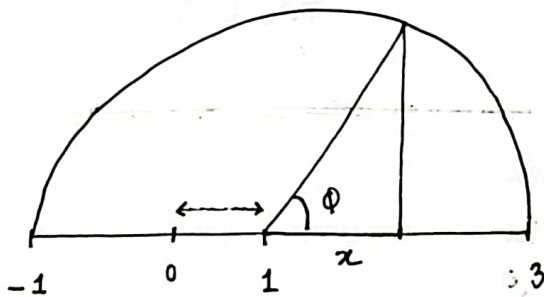
$$x_1 = 1 \cos \frac{(2 \times 1 + 1) \pi}{2(3+1)} = \cos \frac{3\pi}{8} = 0.38$$

$$x_2 = 1 \cos \frac{(2 \times 2 + 1) \pi}{2(3+1)} = \cos \frac{5\pi}{8} = -0.38$$

$$x_3 = 1 \cos \frac{(2 \times 3 + 1) \pi}{2(3+1)} = \cos \frac{7\pi}{8} = -0.92$$

For interval $\rightarrow [-1, 3]$

mid point = $\frac{-1+3}{2} = 1$



radius, $A = 2$, center = 1

here, x কে 0 থেকে count
করতে হয় সাহসু তাই +1 করা
লাগবে

For $[-3, 1]$

radius, $A = 2$, center = -1

shift : center এর position এর উপর base করে x এর সাথে plus - minus করতে হবে

$$\therefore x = A \cos \frac{(2j+1) \pi}{2(n+1)} + \text{center}$$

• Hermite Interpolation

$(n+1)$ nodes $\rightarrow n \rightarrow P_{2n+1}(x)$
 $\searrow$
 degree

3 nodes $\rightarrow n=2 \rightarrow P_{5}(x)$

$(n+1)$ nodes $\rightarrow (n+1)$ conditions
 $\downarrow$
 Derivatives $\rightarrow (n+1)$ conditions
 $\hline$
 $2n+2$ conditions $\rightarrow (2n+2-1)$ degree
 $= (2n+1)$ degree

* More accuracy because degree is increased.

$$P_{2n+1} = \sum_{k=0}^n f(x_k) h_k(x) + f'(x_k) \hat{h}_k(x)$$

hermite basis

$$h_k(x) = [1 - 2(x - x_k) l'_k(x_k)] l_k^2(x)$$

$$\hat{h}_k(x) = (x - x_k) l_k^2(x)$$

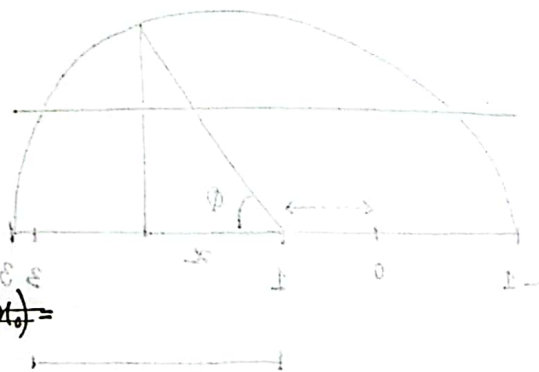
$$\# f(x) = \sin x$$

$$x_0 = 0 \quad x_1 = \pi/2$$

$$f(x_0) = 0 \quad f(x_1) = 1$$

$$f'(x_0) = 1 \quad f'(x_1) = 0$$

$$f'(x_0) = 1$$



$$P_3(x) = f(x_0) h_0(x) + f'(x_0) \hat{h}_0(x) + f(x_1) h_1(x) + f'(x_1) \hat{h}_1(x)$$

$$= 0 + 1 \times \hat{h}_0(x) + 1 \times h_1(x) + 0$$

$$h_1(x) = [1 - 2(x - x_1) l'_1(x_1)] l_1^2(x)$$

$$= [1 - 2(x - \pi/2) \frac{2}{\pi}] \left(\frac{2x}{\pi}\right)^2$$

$$= \left[2 - \frac{4x}{\pi}\right] \frac{4x^2}{\pi^2}$$

$$\hat{h}_0(x) = (x - x_0) l_0^2(x)$$

$$= (x - 0) \left(1 - \frac{2x}{\pi}\right)^2$$

$$l_1(x) = \frac{x - x_0}{x_1 - x_0} = \frac{x - 0}{\pi/2 - 0} = \frac{2x}{\pi}$$

$$l'_1(x) = \frac{2}{\pi}$$

$$l'_1(\pi/2) = \frac{2}{\pi}$$

$$l_0(x) = \frac{x - x_1}{x_0 - x_1} = \frac{x - \pi/2}{0 - \pi/2} = 1 - \frac{2x}{\pi}$$

$$\therefore P_3(x) = x \left(1 - \frac{2x}{\pi}\right)^2 + \left[1 - 2\left(x - \frac{\pi}{2}\right) \frac{2}{\pi}\right] \frac{4x^2}{\pi^2}$$

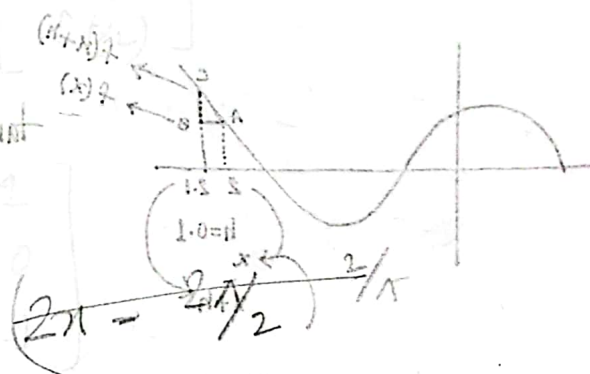
$$\therefore P_3(0) =$$

$$\frac{f(x) - f(x+h)}{h} = \frac{BC}{AB} = \tan \theta$$

$$2 \left(1 - \frac{2x}{\pi}\right)$$

$$1 - \left(2 - \frac{4x}{\pi}\right)$$

$$\Rightarrow 1 - \left(\frac{4x}{\pi} - 2\right)$$



3. Backward Difference

$$\frac{f(x) - f(x-h)}{h} = f'(x)$$

3. Central Difference

$$\frac{f(x+h) - f(x-h)}{2h} = f'(x)$$

[least amount of error so if the better one]

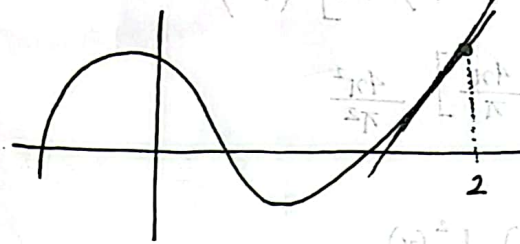
Error of H^2

chapter 3: Differentiation

$$f(x) = x^3 - 4x + 1$$

$$f'(x) = 3x^2 - 4$$

$$f'(2) = 8$$

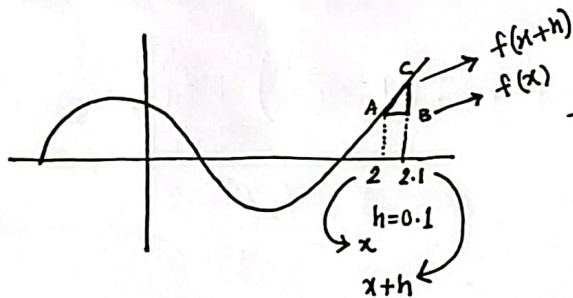


Formulas

1. Forward Difference

$$f'(x) = \frac{f(x+h) - f(x)}{h}$$

$h = \text{step size}$
smaller the h accurate the value of $f'(x)$



$$\tan \theta = \frac{BC}{AB} = \frac{f(x+h) - f(x)}{h}$$

$$\text{error} \propto h$$

2. Backward Difference

$$f'(x) = \frac{f(x) - f(x-h)}{h}$$

3. Central Difference

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} \rightarrow \left[\begin{array}{l} \text{least amount of error} \\ \text{so it the better one} \end{array} \right]$$

$$\text{error} \propto h^2$$

$\# f(x) = \ln(x)$
 find $f'(2) = ?$ using forward diff, central diff
 $h = 1, 0.1, 0.01, 0.001$

Truncation Error = ? $\text{Truncation Error} = \left| \text{Actual} - \text{numeric} \right|$

$f'(x) = \frac{1}{x}$
 $f'(2) = \frac{1}{2} = 0.5 \rightarrow \text{Actual Value}$

h	FD $f'(x)$	FD Truncation Error	CD $f'(x)$	CD Truncation Error	BD $f'(x)$	BD Truncation Error
1	0.4055	0.0945	0.0593	0.0493		
0.1	0.4879	0.0121	0.5004	0.0004		
0.01	0.4988	0.0012	0.5000	0		
0.001	0.4999	0.0001	0.5000	0		

$$\begin{aligned}
 \text{FD} &= \frac{f(2+1) - f(2)}{1} = \frac{\ln(3) - \ln(2)}{1} = 0.4055 \\
 &= \frac{f(2+0.1) - f(2)}{0.1} = \frac{\ln(2.1) - \ln(2)}{0.1} = 0.4879 \\
 &= \frac{f(2+0.01) - f(2)}{0.01} = \frac{\ln(2.01) - \ln(2)}{0.01} = 0.4988 \\
 &= \frac{f(2+0.001) - f(2)}{0.001} = \frac{\ln(2.001) - \ln(2)}{0.001} = 0.4999
 \end{aligned}$$

$$\text{error}_1 = 0.5 - 0.4055 = 0.0945$$

$$\text{error}_2 = 0.5 - 0.4879 = 0.0121$$

$$\text{error}_3 = 0.5 - 0.4988 = 0.0012$$

$$\text{error}_4 = 0.5 - 0.4999 = 0.0001$$

$$\begin{aligned} \bar{CD} &= \frac{f(2+1) - f(2-1)}{2 \times 1} = \frac{\ln(3) - \ln(1)}{2} = 0.5493 \\ &= \frac{f(2+0.1) - f(2-0.1)}{(2 \times 0.1)} = \frac{\ln(2.1) - \ln(1.9)}{0.2} = 0.5004 \\ &= \frac{f(2+0.01) - f(2-0.01)}{(2 \times 0.01)} = \frac{\ln(2.01) - \ln(1.99)}{0.02} = 0.5000 \\ &= \frac{f(2+0.001) - f(2-0.001)}{(2 \times 0.001)} = \frac{\ln(2.001) - \ln(1.999)}{2 \times 10^{-3}} = 0.5000 \end{aligned}$$

$$\text{error}_1 = 0.5 - 0.5493 = -0.0493$$

$$\text{error}_2 = 0.5 - 0.5004 = -0.0004$$

$$\text{error}_3 = 0.5 - 0.5000 = 0$$

$$\text{error}_4 = 0.5 - 0.5000 = 0$$

h	f(h)	FD Error	FD
1	0.4052	0.0493	0.5493
0.1	0.4244	0.0121	0.5004
0.01	0.4482	0.0012	0.5000
0.001	0.4444	0.0001	0.5000

$$FD = \frac{f(2+1) - f(2-1)}{1} = \frac{\ln(3) - \ln(1)}{1} = 0.4052$$

$$FD = \frac{f(2+0.1) - f(2-0.1)}{0.2} = \frac{\ln(2.1) - \ln(1.9)}{0.2} = 0.4244$$

$$FD = \frac{f(2+0.01) - f(2-0.01)}{0.02} = \frac{\ln(2.01) - \ln(1.99)}{0.02} = 0.4482$$

$$FD = \frac{f(2+0.001) - f(2-0.001)}{0.002} = \frac{\ln(2.001) - \ln(1.999)}{0.002} = 0.4444$$

$$\begin{aligned} \text{error}_1 &= 0.5 - 0.4052 = 0.0448 \\ \text{error}_2 &= 0.5 - 0.4244 = 0.0756 \\ \text{error}_3 &= 0.5 - 0.4482 = 0.0518 \\ \text{error}_4 &= 0.5 - 0.4444 = 0.0556 \end{aligned}$$

$f(x) = x^3 + 2$

find, $f'(1.8)$ using FD.

x_0	x_1	x_2
1.2	1.5	1.8

$h = 1.8 - 1.5$
 $= 0.3$

$\therefore f'(1.8) = \frac{f(x+h) - f(x)}{h}$

$= \frac{f(1.8 + 0.3) - f(1.8)}{0.3} = \frac{f(2.1) - f(1.8)}{0.3}$

$= \frac{11.261 - 7.832}{0.3} = 11.43$

$f(x) = x^3 + 2$

find $f'(1.5)$ using CD.

x_0	x_1	x_2
1.2	1.5	1.8
$f(x) \rightarrow$ 10	15	25

$h = 1.8 - 1.5$
 $= 0.3$

$\therefore f'(1.5) = \frac{f(x+h) - f(x-h)}{2h}$

$= \frac{f(1.5 + 0.3) - f(1.5 - 0.3)}{2 \times 0.3}$

$= \frac{f(1.8) - f(1.2)}{2 \times 0.3}$

$= \frac{25 - 10}{0.6}$

$= 25$

• Truncation Error

• Forward Difference / Backward Difference

$$f(x) = P_1(x) + \text{Error}$$

$$f(x) = f(x_0) L_0(x) + f(x_1) L_1(x) + \text{Error}$$

$$f(x) = \frac{x-x_1}{x_0-x_1} f(x_0) + \frac{x-x_0}{x_1-x_0} f(x_1)$$

$$+ \frac{f''(\xi)}{2!} (x-x_0)(x-x_1)$$

$$f'(x) = \frac{f(x_1) - f(x_0)}{x_1 - x_0} + \frac{f''(\xi)}{2!} (x_0 - x_1)$$

$$= \frac{f(x_0+h) - f(x_0)}{x_0+h-x_0} + \frac{f''(\xi)}{2!} (x_0 - x_0-h)$$

$$= \frac{f(x_0+h) - f(x_0)}{h} + \frac{f''(\xi)}{2} (-h)$$

$$\therefore \text{Truncation error} = \left| \frac{f''(\xi)}{2!} (-h) \right|$$

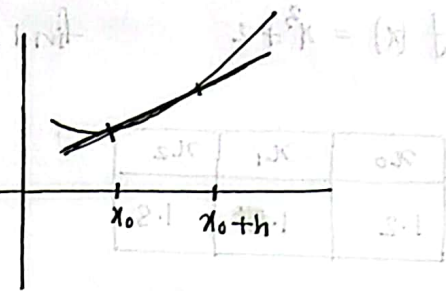
$$= \left| \frac{f''(\xi)}{2!} h \right| \rightarrow \text{forward and backward difference are same}$$

• Central difference

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} - \frac{f'''(\xi)}{3!} h^2$$

$$\therefore \text{Truncation error} = \left| -\frac{f'''(\xi)}{3!} h^2 \right|$$

$$= \frac{f'''(\xi)}{3!} h^2$$



2 nodes

1 degree

x_0

$x_1 = x_0+h$

$P_1(x)$

x_0	x_0+h	x_1
$f(x_0)$	$f(x_0+h)$	$f(x_1)$
h	error	

$$\frac{(x-x_1)f(x_0) - (x-x_0)f(x_1)}{x_0-x_1} \rightarrow \text{error}$$

3 nodes

2 degree

x_0-h

x_0

x_0+h

$P_2(x)$

Central difference

Total error = Truncation error + Rounding error

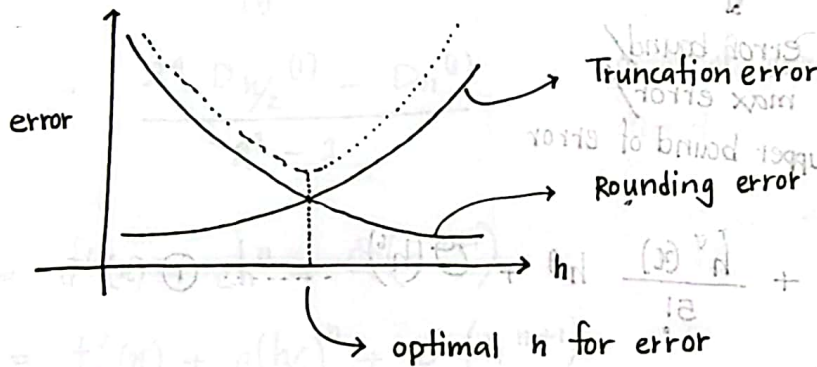
$$= \frac{f'''(\xi)}{3!} h^2 + \epsilon_m \frac{f(x+h) - f(x-h)}{2h}$$

optimization $\rightarrow$ Truncation $\hookrightarrow$ $h \uparrow$ error $\uparrow$
 $h \downarrow$ error $\downarrow$

But Rounding $\hookrightarrow$

$h \uparrow$ error $\downarrow$
 $h \downarrow$ error $\uparrow$

Thus we do optimization.



Richardson Extrapolation

$$D_h = f'(x) = \frac{f(x+h) - f(x-h)}{2h}$$

$$D_{h/2} = \frac{f(x+h/2) - f(x-h/2)}{2 \cdot h/2}$$

Taylor series expansion $\rightarrow$

$$f(x_1+h) = f(x_1) + f'(x_1)h + \frac{f''(x_1)}{2!}h^2 + \frac{f'''(x_1)}{3!}h^3 + \dots$$

centering at x_1

$$f(x_1+h) = f(x_1) + f'(x_1)h + \frac{f''(x_1)}{2!}h^2 + \frac{f'''(x_1)}{3!}h^3 + \frac{f^{(4)}(x_1)}{4!}h^4 + \frac{f^{(5)}(x_1)}{5!}h^5 + O(h^6)$$

$$f(x_1 - h) = f(x_1) - f'(x_1)h + \frac{f''(x_1)}{2!}h^2 - \frac{f'''(x_1)}{3!}h^3 + \frac{f^{(4)}(x_1)}{4!}h^4 - \frac{f^{(5)}(x_1)}{5!}h^5 + O(h^6)$$

$$\therefore f(x_1 + h) - f(x_1 - h) = 2f'(x_1)h + 2\frac{f'''(x_1)}{3!}h^3 + 2\frac{f^{(5)}(x_1)}{5!}h^5 + O(h^7)$$

$$\underbrace{\frac{f(x_1 + h) - f(x_1 - h)}{2h}}_{\text{central difference formula}} = f'(x_1) + \underbrace{\left[\frac{f'''(x_1)}{3!}h^2 + \frac{f^{(5)}(x_1)}{5!}h^4 + O(h^6) \right]}_{\text{Truncation error expansion}}$$

doesn't matter if we multiply anything so 2
 लिखावा ना but matters when we multiply or divide with h.

extrapolation

$$D_h = f'(x) + \frac{f'''(x)}{3!}h^2 + \frac{f^{(5)}(x)}{5!}h^4 + O(h^6) \quad \text{--- (1)}$$

$$D_{h/2} = f'(x) + \frac{f'''(x)}{3!}\frac{h^2}{4} + \frac{f^{(5)}(x)}{5!}\frac{h^4}{16} + O(h^6) \quad \text{--- (2)}$$

$$\text{(2)} \times 4 - \text{(1)}$$

$$4D_{h/2} - D_h = 3f'(x) + 0 + \left(\frac{1}{4} - 1\right)\frac{f^{(5)}(x)}{5!}h^4 + O(h^6)$$

$$\frac{4D_{h/2} - D_h}{3} = f'(x) - \frac{1}{4}\frac{f^{(5)}(x)}{5!}h^4 + O(h^6)$$

$$\therefore D_h^{(1)} = \frac{4D_{h/2} - D_h}{3} = \frac{2^2 D_{h/2} - D_h}{2^2 - 1} \quad \text{order of error } n=2$$

h^2 - degree 2 error / 2nd order of error
 h^4 - degree 4 error / 4th order of error



$$D_h^{(1)} = f'(x) - \frac{1}{4} \frac{f''(x)}{5!} h^4 + O(h^6) \dots \dots \dots \textcircled{1}$$

$$D_{h/2}^{(1)} = f'(x) - \frac{1}{4} \frac{f''(x)}{5!} \frac{h^4}{16} + O(h^6) \dots \dots \dots \textcircled{2}$$

$$\textcircled{2} \times 16 - \textcircled{1}$$

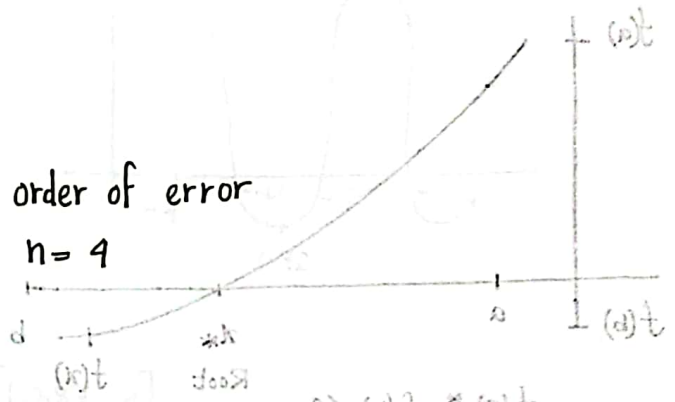
$$16 D_{h/2}^{(1)} - D_h^{(1)} = 15 f'(x) + 0 + O(h^6)$$

$$\frac{16 D_{h/2}^{(1)} - D_h^{(1)}}{15} = f'(x) + O(h^6)$$

$$D_h^{(2)} = \frac{16 D_{h/2}^{(1)} - D_h^{(1)}}{15}$$

$$= \frac{2^4 D_{h/2}^{(1)} - D_h^{(1)}}{2^4 - 1}$$

order of error
n = 4

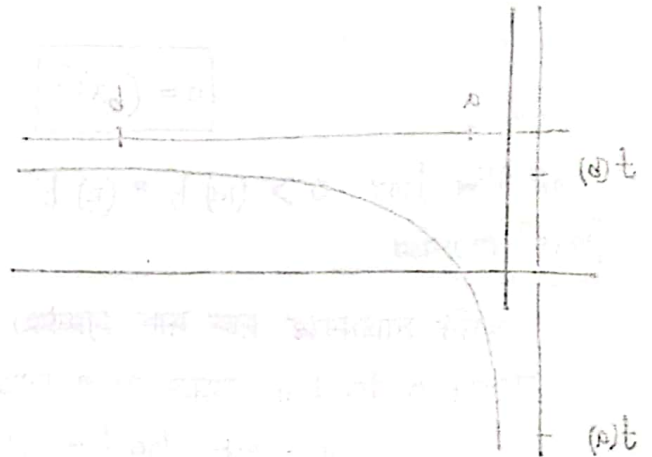


$$D_h = f'(x) + c h^n + O(h^{n+1})$$

$$D_{h/2} = f'(x) + c (h/2)^n + O(h^{n+1})$$

$$D_h^{(1)} = \frac{2^n D_{h/2} - D_h}{2^n - 1}$$

$$D_h^{(2)} = \frac{2^n D_{h/2}^{(1)} - D_h^{(1)}}{2^n - 1}$$



$f(a) > 0, f(b) < 0$

$f(a) < 0, f(b) > 0$

if $f(a) > 0$ and $f(b) < 0$ then root lies between a and b

if $f(a) < 0$ and $f(b) > 0$ then root lies between a and b

if $f(a) = 0$ or $f(b) = 0$ then root lies either at a or b

chapter 4 - Non linear Equation

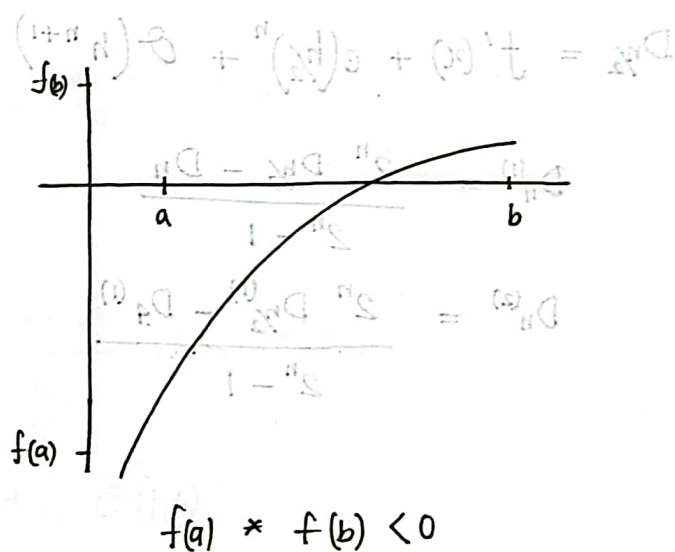
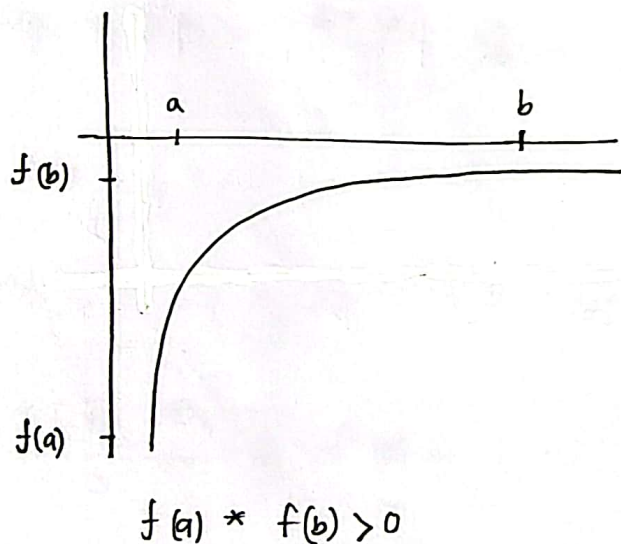
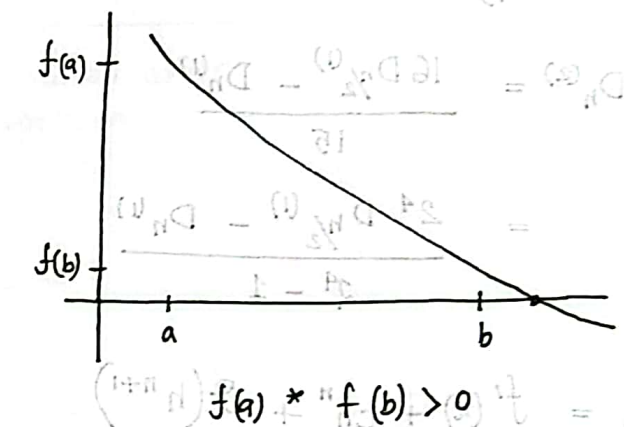
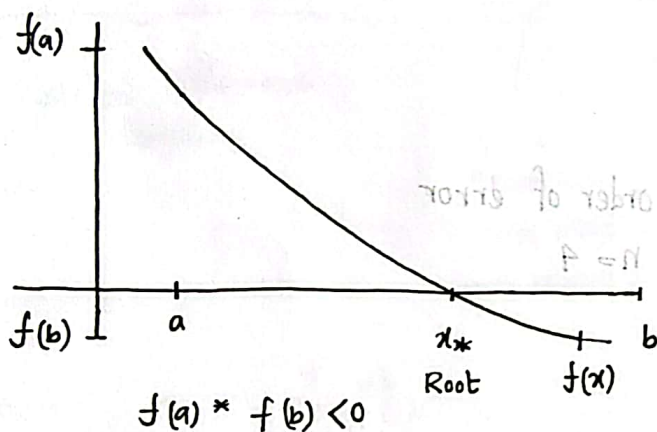
$$x^2 - 2x = 0$$

$$\Rightarrow x(x-2) = 0$$

$$\Rightarrow x = 0, 2$$

Root finding iterative algorithm

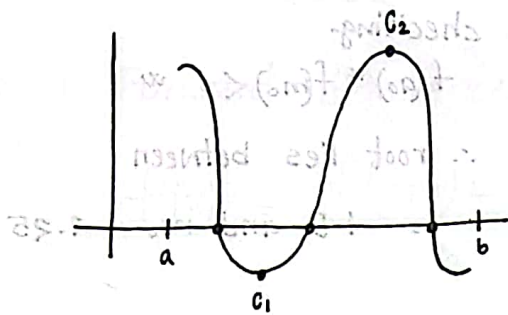
Bisection method



$\therefore f(a) * f(b) < 0 \rightarrow$ root lies between a and b

$\therefore f(a) * f(b) > 0 \rightarrow$ root does not lie between a and b

$\therefore f(a) * f(b) = 0 \rightarrow$ root lies either on a or b



This method can only deal with one root in an interval. We need to split the interval so that at a time there is only one root in an interval.

We will use turning points to divide intervals

[আরা ছেদ করে না তাদের ignore]

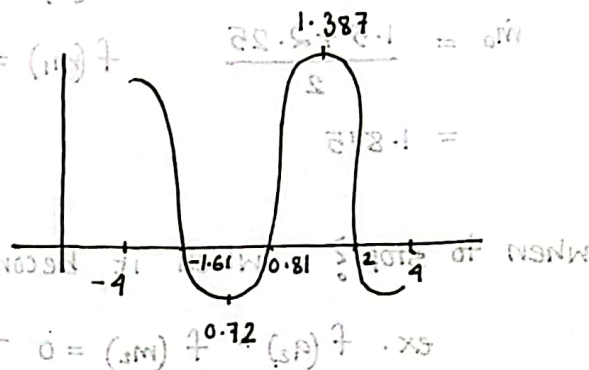
$$[a, b] = [a, c_1] \cup [c_1, c_2] \cup [c_2, b]$$

$$\# f(x) = x^3 - x^2 - 3x + 2$$

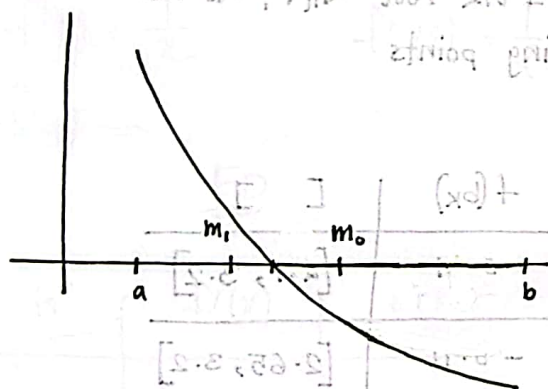
$$[-4, 4]$$

roots are 2, 0.81, -1.61

turning points 0.72, 1.387



$$[a, b] = [-4, 0.72] \cup [0.72, 1.387] \cup [1.387, 4]$$



$$f(x_*) = 0$$

$f(a) * f(m) < 0$ root will lie between $[a, m]$

এইভাবে বার-বার স্কাইস্ক্যান দিয়ে

ভাঙা করতে থাকবে and at a point we will get the root.

$$\# f(x) = \frac{1}{x} - 0.5$$

Interval $[1.5, 3]$

Actual root = 2

find actual root

$$\frac{1}{x} - 0.5 = 0$$

$$\Rightarrow x = 2$$

$K=0$ $a_0 = 1.5$ $f(a_0) = 0.166$
 $b_0 = 3$ $f(b_0) = -0.166$

checking

$f(a_0) * f(b_0) < 0$ ✓

∴ root lies between

$a_0 = 1.5$ and $b_0 = 3$

$m_0 = \frac{a_0 + b_0}{2}$ $f(m_0) = -0.055$
 $= 2.25$

[Bisection Method]

$K=1$ $a_1 = 1.5$ $f(a_1) = 0.166$
 $b_1 = 2.25$ $f(b_1) = -0.055$
 $m_1 = \frac{1.5 + 2.25}{2}$ $f(m_1) = 0.033$
 $= 1.875$

checking $[1.5, 2.25] = [a, b]$

$f(a_1) * f(b_1) > 0$

∴ root lies between

$m_1 = 1.875$ and $b_1 = 2.25$

when to stop? - when it becomes "0"

ex. $f(a_2) * f(m_2) = 0$ অর্থ root found

✓* $f(x) = x^3 - 7x^2 + 14x - 6 = 0$

interval $\rightarrow [1, 3.2]$

একটি interval দেয়া থাকবে

যা 1 টি root থাকে, No need for turning points

Accurate within/error bound/ $\epsilon_m = 10^{-3}$

means $|f(m_k)| \leq 10^{-3}$ অর্থাৎ যখন যাবে

[K]	a_k	m_k	b_k	$f(a_k)$	$f(m_k)$	$f(b_k)$	[]
0	1	2.1	3.2	2	1.79	-0.11	[2.1, 3.2]
1	2.1	2.65	3.2	1.79	0.55	-0.11	[2.65, 3.2]
2	2.65	2.925	3.2	0.55	0.086	-0.11	[2.925, 3.2]
3	2.925	3.065	3.2	0.086	-0.054	-0.11	[2.925, 3.065]
...	...	...	...	...	...	...	...
9	2.998...	3.000195	3.002	1.96×10^{-3}	-1.95×10^{-4}	-2.3×10^{-3}	$\epsilon = 10^{-3}$

∴ Approximate root = 3.000195

Problems of bisection method

- ① More iterations needed to find root
- ② Slow converge rate

Minimum number of iterations to find root

$$n \geq \frac{\log(|b-a|) - \log(\epsilon)}{\log(2)}$$

interval $[a, b]$

$\epsilon \rightarrow$ error bound / machine epsilon / accuracy

$$[1.5, 3]$$

$$\epsilon = 1.1 \times 10^{-16}$$

$$n \geq \frac{\log(|3-1.5|) - \log(1.1 \times 10^{-16})}{\log(2)}$$

$$n \geq 52.59$$

$$n \geq 53 \text{ [taking the ceiling value]}$$

converge

→ দিকে যাওয়া

diverge

→ দূরে আরে যাওয়া

here, log base is 10 ($\log_{10}$)

$$\frac{\epsilon + \kappa \epsilon}{2} = \kappa \epsilon$$

$$(10) \epsilon = \kappa \epsilon$$

$$\left\{ \begin{array}{l} \kappa = 0 \\ \frac{\epsilon + \kappa \epsilon}{2} = 0 \end{array} \right.$$

$$\frac{\epsilon + \kappa \epsilon}{2} = (10) \epsilon \quad (1)$$

ceiling value

$$n \geq 7.001$$

$$n \geq 8$$

$$P2.5 = (P2.5) \epsilon$$

$$EP.5 = (EP.5) \epsilon$$

$$AP.5 = (AP.5) \epsilon$$

$$PP.5 = (PP.5) \epsilon$$

$$OP.5 = (OP.5) \epsilon$$

$$\epsilon = (10) \epsilon$$

converge

converge

$$\epsilon = 1000$$

Fixed Point Iteration

$$f(x) = x^2 - 2x - 3 = 0$$

$$\rightarrow x_* = -1, 3 \text{ (roots)}$$

More iterations used to find root

down arrow
converges faster

$$x^2 - 2x - 3 = 0$$

$$\Rightarrow x^2 = 2x + 3$$

$$\Rightarrow x = \sqrt{2x+3}$$

① $g(x)$

$$\therefore x = g(x)$$

$$y = x$$

$$y = \sqrt{2x+3}$$

plotting these two will
intercept at the root
which is 3.

$$x^2 - 2x - 3 = 0$$

$$\Rightarrow x^2 - x - x - 3 = 0$$

$$\Rightarrow x = \frac{x^2 - x - 3}{x - 1}$$

② $g(x)$

$$\Rightarrow 2x^2 - x^2 - 2x - 3 = 0$$

$$\Rightarrow x(2x-2) = x^2 + 3$$

③ $g(x)$

$$\textcircled{I} \quad g(x) = \sqrt{2x+3}, \quad x_0 = 0, 42$$

$$\textcircled{II} \quad g(x) = \frac{x^2 - x - 3}{x - 1}, \quad x_0 = 0, 42$$

$$g(0) = 1.73$$

$$g(0) = -3$$

$$g(1.73) = 2.54$$

$$g(-3) = 9$$

$$g(2.54) = 2.84$$

$$g(9) = 69$$

$$g(2.84) = 2.95$$

$$g(2.95) = 2.98$$

$$g(2.98) = 2.99$$

$$g(2.99) = 3$$

$$g(3) = 3$$

Fixed point
reached
root = 3

converging
towards
root 3

$$g(12) = 1.72 \times 10^3$$

$$g(1.72 \times 10^3) = 2.95 \times 10^6$$

diverges

③ $g(x) = \frac{x^2 + 3}{2x - 2}$, $x_0 = 0, 42$

$g(0) = -1.5$

$g(-1.5) = -1.05$

$g(-1.05) = -1$

$g(-1) = -1$

fixed point
reached
root = -1

converging
towards
root

$g(42) = 21.6$

$g(21.6) = 11.4$

$g(11.4) = 6.39$

$g(6.39) = 4.07$

$g(4.07) = 3.19$

$g(3.19) = 3.01$

$g(3.01) = 3$

$g(3) = 3$

fixed point
reached

root = 3

converging
towards
root

① $g(42) = 9.33$

$g(9.33) = 4.65$

$g(4.65) = 3.51$

⋮

$g(3) = 3$

fixed point
reached
root = -1

converging
towards
root

Contraction Mapping Theory

Find $g'(x)$

Find $|g'(\text{root})|$

convergence rate, $\lambda = |g'(\text{root})|$

order of convergence $\left\{ \begin{array}{l} 0 < \lambda < 1 \longrightarrow \text{linear convergence (will converge)} \\ \lambda = 0 \longrightarrow \text{super linear convergence (best as fast \& iteration converge)} \\ \lambda = 1 \longrightarrow \text{ignore} \\ \lambda > 1 \longrightarrow \text{diverge} \end{array} \right.$



converging towards

converging towards

$$① \quad g(x) = \sqrt{2x+3}$$

$$\lambda = |g'(-1)| = 1 \quad (\text{ignore})$$

$$\lambda = |g'(3)| = 1/3 \quad (\text{linear convergence})$$

$$② \quad g(x) = x^2 - x + 3$$

$$\lambda = |g'(-1)| = 3 \quad (\text{diverge})$$

$$\lambda = |g'(3)| = 5 \quad (\text{diverge})$$

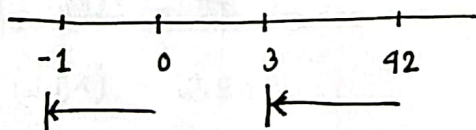
$$③ \quad g(x) = \frac{x^2+3}{2x-2}$$

$$\lambda = |g'(-1)| = 0 \quad (\text{super linear convergence})$$

$$\lambda = |g'(3)| = 0 \quad (\text{super linear convergence})$$

$$0 \longrightarrow -1$$

$$42 \longrightarrow 3$$



converging
towards

converging
towards

$$|g'(x)| = \lambda$$

$$0 < \lambda < 1 \longrightarrow \text{linear convergence (will converge)}$$

$$\lambda = 0 \longrightarrow \text{super linear convergence (fast as fast as possible)}$$

$$\lambda = 1 \longrightarrow \text{ignore}$$

$$\lambda > 1 \longrightarrow \text{diverge}$$

convergence

Newton's Method

- it's a fixed point iteration method

- it is super linear convergent usually ($\lambda = 0$)

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$$\lambda = |g'(x)| = \left| \frac{f(x) f''(x)}{[f'(x)]^2} \right|$$

at root x_*
 $f(x) = 0$ so $\lambda = 0$.

$f(x) = x^2 - 2xe^{-x} + e^{-2x}$, $x_0 = 1$, error bound, $\epsilon_m = 0.00001$

$$f'(x) = 2x - 2e^{-x} - 2x(-e^{-x}) - 2e^{-2x}$$

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$$= x_k - \frac{x_k^2 - 2x_k e^{-x_k} + e^{-2x_k}}{2x_k - 2e^{-x_k} + 2x_k(-e^{-x_k}) - 2e^{-2x_k}}$$

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$ $\therefore$ Approx Root = 0.568615

K	x_k	$f(x_k)$	$ f(x_k) < 0.0001$
0	1	0.399576	No
1	0.768941	0.093292	No
2	0.664590	0.022532	No
...	...	...	...
8	0.568615	0.5×10^{-5}	Yes

Problem of Newton's Method

When $f'(x) = 0$ we don't get the value of $g(x)$

↳ derivative 0 in turning points case.

Solution to this → Secant Method / Quasi-Newton's Method

$$f'(x_k) = \frac{f(x) - f(x-h)}{h} \quad (\text{doing backward difference})$$

$$= \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$= \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}$$

Problem :

Initial value

$$\therefore x_{k+1} = x_k - \frac{f(x_k)(x_k - x_{k-1})}{f(x_k) - f(x_{k-1})}$$

$$\# f(x) = \frac{1}{x} - 0.5, \quad x_0 = 0.25, \quad x_1 = 0.5$$

K	x_k	$f(x_k)$
0	0.25	
1	0.5	
2	0.6875	
3	1.01562	
⋮	⋮	⋮
12	2	0

The helping method to help with the speed of fixed point iteration, Newton's method, secant method →

Aitken Method

Not a method itself but works as a helper

$$\# f(x) = \frac{1}{x} - 0.5, \quad x_0 = 1.5$$

$$g(x) = x + \frac{1}{16} \left(\frac{1}{x} - 0.5 \right)$$

Quiz 4

Assignment 4

31st March

$$x_0 = 1.5$$

$$x_1 = g(x_0) = 1.510417$$

$$x_2 = g(x_1) = 1.520546$$

$$x_3 = g(x_2) = 1.530400$$

⋮

$$x_{818} = 2$$

$$\hat{x}_{k+2} = x_k - \frac{(x_{k+1} - x_k)^2}{x_{k+2} - 2x_{k+1} + x_k}$$

প্রতি 2 iteration পর পর
Aitken method apply
করা যায়

$$x_0 = 1.5$$

$$x_1 = g(x_0) = 1.510417$$

$$x_2 = g(x_1) = 1.520546$$

$$\hat{x}_2 = x_0 - \frac{(x_1 - x_0)^2}{x_2 - 2x_1 + x_0} = 1.877604 \quad (\text{new } x_2 \rightarrow \text{আগের } x_2 \text{ বাদ})$$

$$x_3 = g(\hat{x}_2) = 1.879641$$

$$x_4 = g(x_3) = 1.881642$$

$$\hat{x}_4 = \hat{x}_2 - \frac{(x_3 - \hat{x}_2)^2}{x_4 - 2x_3 + \hat{x}_2} = 1.992683 \rightarrow x_5 \rightarrow x_6 \rightarrow \hat{x}_6 \rightarrow x_7 \rightarrow x_8 \rightarrow \hat{x}_8 = 2$$

chapter 5 : Linear Equations

$$x_1 + 2x_2 + x_3 = 0$$

$$x_1 - 2x_2 + 2x_3 = 4$$

$$2x_1 + 12x_2 - 2x_3 = 4$$

$$\underbrace{\begin{pmatrix} 1 & 2 & 1 \\ 1 & -2 & 2 \\ 2 & 12 & -2 \end{pmatrix}}_a \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}}_x = \underbrace{\begin{pmatrix} 0 \\ 4 \\ 4 \end{pmatrix}}_b$$

$$Ax = b$$
$$\Rightarrow x = A^{-1} b$$

Inverse matrix conditions :

- ① Square matrix
- ② $\text{Det}(A) \neq 0$ [Non-singular matrix]

The methods computer follows to inverse a matrix :

- ① Gaussian Elimination
- ② LU Decomposition

Gaussian Elimination

3x3 upper triangular

$$\begin{pmatrix} 2 & 3 & 1 \\ 0 & 6 & 4 \\ 0 & 0 & 5 \end{pmatrix}$$

backward
substitution

primary
diagonal

3x3 lower triangular

$$\begin{pmatrix} 5 & 0 & 0 \\ 1 & 15 & 0 \\ 10 & 2 & 3 \end{pmatrix}$$

forward
substitution

primary
diagonal

Augmented matrix

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 1 & -2 & 2 & 4 \\ 2 & 12 & -2 & 4 \end{array} \right)$$

sequence $21 \rightarrow 31 \rightarrow 32$

then $21 \rightarrow 31 \rightarrow 32$

$$\text{Row multiplier, } m_{ik} = \frac{a_{ik}}{a_{kk}}$$

$$\text{Row multiplier, } m_{21} = \frac{a_{21}}{a_{11}} = \frac{1}{1} = 1$$

$$r'_2 = r_2 - (r_1 \times m_{21})$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -4 & 1 & 4 \\ 2 & 12 & -2 & 4 \end{array} \right)$$

$$\text{Row multiplier, } m_{31} = \frac{a_{31}}{a_{11}} = \frac{2}{1} = 2$$

$$r'_3 = r_3 - (r_1 \times m_{31})$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -4 & 1 & 4 \\ 0 & 8 & -4 & 4 \end{array} \right)$$

$$\text{Row multiplier, } m_{32} = \frac{a_{32}}{a_{22}} = -2$$

$$r'_3 = r_3 - (r_2 \times m_{32})$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -4 & 1 & 4 \\ 0 & 0 & -2 & 12 \end{array} \right)$$

backward substitution,

$$-2x_3 = 12$$

$$\therefore x_3 = -6$$

$$-4x_2 + x_3 = 4$$

$$\Rightarrow -4x_2 = 10$$

$$\therefore x_2 = -\frac{5}{2}$$

$$x_1 + 2x_2 + x_3 = 0$$

$$\Rightarrow x_1 - 5 - 6 = 0$$

$$\therefore x_1 = 11$$

Number of operations of backward/forward substitution

computational complexity

$$n^2$$

Dimension

Number of operations of Gaussian Elimination

computational complexity

$$\frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n$$

LU Decomposition $\longrightarrow$ better because of reusability

for different values of b Gaussian needs to do from the very start but in LU we can just change the b later.

$$A = LU$$

$$Ax = b$$

$$\Rightarrow LUx = b$$

$$Lx = y$$

$$Ly = b$$

$y = ? \longrightarrow$ using forward substitution calc the y

$x = ? \longrightarrow$ using backward substitution calc the x

$$\# \quad x_1 + 2x_2 + x_3 = 0$$

$$x_1 - 2x_2 + 2x_3 = 0$$

$$2x_1 + 12x_2 - 2x_3 = 4$$

$$3 \times 3 \quad \begin{pmatrix} 1 & 2 & 1 \\ 1 & -2 & 2 \\ 2 & 12 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix}$$

$A \quad x \quad b$

Frobinous matrix $\longrightarrow$ unit lower triangular matrix

$$F^{(1)} = \begin{pmatrix} 1 & 0 & 0 \\ -m_{21} & 1 & 0 \\ -m_{31} & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$$

$$m_{21} = \frac{a_{21}}{a_{11}} = \frac{1}{1} = 1$$

$$m_{31} = \frac{a_{31}}{a_{11}} = \frac{2}{1} = 2$$

for different values of b $A^{(2)} = F^{(2)} \times A$

$$= \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 1 & -2 & 2 \\ 2 & 12 & -2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & 1 \\ 0 & 8 & -4 \end{pmatrix}$$

$$A = LU$$

$$Ax = b$$

$$L(Ux) = b$$

$$Ux = y$$

$$L^{-1}b = y$$

$$y = ?$$

$$x = ?$$

$$F^{(2)} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -m_{32} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}$$

$$m_{32} = \frac{a_{32}}{a_{22}} = \frac{8}{-4} = -2$$

$$A^{(3)} = F^{(2)} \times A^{(2)}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & 1 \\ 0 & 8 & -4 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & 1 \\ 0 & 0 & -2 \end{pmatrix} = U$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ m_{21} & 1 & 0 \\ m_{31} & m_{32} & 1 \end{pmatrix}$$

unit lower triangular matrix

$$= \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -2 & 1 \end{pmatrix}$$

$$F^{(1)} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

$$m_{21} = \frac{a_{21}}{a_{11}} = \frac{1}{1} = 1$$

$$m_{31} = \frac{a_{31}}{a_{11}} = \frac{2}{1} = 2$$

$$Ly = b$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -2 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \\ 4 \end{pmatrix}$$

prüfung

Forward substitution →

$$y_1 = 0$$

$$y_1 + y_2 = 4$$

$$\therefore y_2 = 4$$

$$2y_1 - 2y_2 + y_3 = 4$$

$$\therefore y_3 = 12$$

$$Ux = y$$

$$\Rightarrow \begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & 1 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \\ 12 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 4 \\ 12 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 12 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 4 \\ 12 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\frac{0}{0} = \frac{12}{12} = 1$$

Backward substitution →

$$-2x_3 = 12$$

$$\therefore x_3 = -6$$

$$-4x_2 + x_3 = 4$$

$$\therefore x_2 = -5/2$$

$$x_1 + 2x_2 + x_3 = 0$$

$$\therefore x_1 = 11$$

$$\begin{pmatrix} 0 \\ 4 \\ 12 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 12 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$\therefore \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 11 \\ -5/2 \\ -6 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 4 \\ 12 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 12 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Pivoting → swapping $\begin{cases} \text{row wise} \\ \text{column wise} \end{cases}$

$$d = \sqrt{2}$$

$$3x_2 = 3$$

$$2x_1 = 2$$

$$x_3 = 1$$

$$\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 2 & 2 \end{pmatrix} \leftarrow$$

pivot point → ক্ষেত্রে ০ থাকে তবে না কোথা ৩

Forward substitution →

$$x_1 = 0$$

$$P = 2x_2 + 1x_3$$

$$P = 2x_2 \therefore$$

$$P = 2x_2 + 2x_3 - 1x_3$$

$$2x_2 = 1x_3$$

$$\begin{pmatrix} 0 & 3 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} 0 & 3 & 0 & 3 \\ 2 & 0 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right)$$

$$m_{21} = \frac{a_{21}}{a_{11}} = \frac{2}{0} \quad [\text{unidentified oh no!}]$$

swapping row 1 and 2 →

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

[suggested]

OR

swapping column 1 and 2 →

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_2 \\ x_1 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$$

unchanged

Chapter 6 - Least Square Approximation

when it is not a square matrix $\rightarrow$ not $n \times n$ but $m \times n$ where $m > n$

number of equations $>$ number of variables

(m) Rows $>$ column (n)

over determined System

Discrete least square

QR Decomposition

Discrete least square method

$$Ax = b$$

$\downarrow$

$$m \times n \quad (m > n)$$

$$A^T Ax = A^T b$$

$\downarrow \quad \downarrow$

$$n \times m \quad m \times n$$

$\downarrow$

$$n \times n$$

$$\therefore x = (A^T A)^{-1} A^T b$$

Fit a straight line using the nodes.

$$P_1(x) = a_0 + a_1 x \quad [n=1]$$

$$x_0 = -3 \quad x_1 = 0 \quad x_2 = 6$$

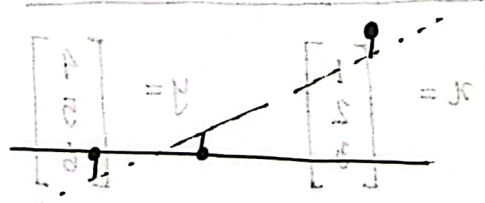
$$f(x_0) = 0 \quad f(x_1) = 0 \quad f(x_2) = 2$$

$$P_1(x_0) = a_0 x_0^0 + a_1 x_0^1 = f(x_0)$$

$$P_1(x_1) = a_0 x_1^0 + a_1 x_1^1 = f(x_1)$$

$$P_1(x_2) = a_0 x_2^0 + a_1 x_2^1 = f(x_2)$$

dot product / inner product



degree n
to node n+1

$$\begin{bmatrix} x_0^0 & x_0^1 \\ x_1^0 & x_1^1 \\ x_2^0 & x_2^1 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} f(x_0) \\ f(x_1) \\ f(x_2) \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ -3 & 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -3 & 0 & 6 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 3 \\ 3 & 45 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 2 \\ 12 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 45 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 12 \end{bmatrix} = \begin{bmatrix} 3/7 \\ 5/21 \end{bmatrix}$$

$$p_1(x) = 3/7 + 5/21 x$$

Orthogonality

2 types of notation $\rightarrow$ vector notation : $\vec{x} \cdot \vec{y} = xy \cos \theta$
 matrix notation : $x^T y$ or $y^T x$ or $(A^T A) = x \dots$

dot product / inner product

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad y = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$x^T y = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = 32$$

$$\vec{x} \cdot \vec{y} = (1 \times 4) + (2 \times 5) + (3 \times 6) = 32$$

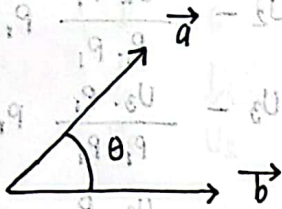
l_2 norm $\rightarrow$ inner product with itself : $x^T x$ or $\vec{x} \cdot \vec{x}$

length / Magnitude

$$a = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$|a| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

Dot Product



if $\theta = 90^\circ$ or $\pi/2$

$$\cos 90^\circ = 0$$

$$\vec{a} \cdot \vec{b} = ab \cos 90^\circ = 0 \longrightarrow \text{orthogonal vectors}$$

$$\therefore \vec{a} \cdot \vec{b} = 0$$

$$\text{or } a^T b = 0$$

$$\# S = \{\vec{a}, \vec{b}, \vec{c}\}$$

S is an orthogonal set if

$$\vec{a} \cdot \vec{b} = 0$$

$$\vec{b} \cdot \vec{c} = 0$$

$$\vec{c} \cdot \vec{a} = 0$$

* orthonormality on the last page

$$\boxed{R^T Q = I} \therefore$$

$$\boxed{R = Q^T A} \therefore$$

chapter 6 - least Square Approximation

Orthonormality

if ① The vectors are orthogonal (dot product = 0)

② The length of the vectors = 1 (unit vectors)

$$\vec{a} = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$$

orthogonality test $\rightarrow$

$$\vec{a} \cdot \vec{b} = (4 \times 1) + (2 \times -3) + (1 \times 2) = 0$$

$\therefore$ orthogonal

unit vector test $\rightarrow$

$$|\vec{a}| = \sqrt{4^2 + 2^2 + 1^2} = \sqrt{21} \rightarrow \text{not unit vector}$$

$$|\vec{b}| = \sqrt{1^2 + (-3)^2 + 2^2} = \sqrt{14} \rightarrow \text{not unit vector}$$

$\therefore \vec{a}$ and $\vec{b}$ are orthogonal but not orthonormal.

Normalization - জোর করে unit vector = 1 বানানোর method

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{1}{\sqrt{21}} \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{4}{\sqrt{21}} \\ \frac{2}{\sqrt{21}} \\ \frac{1}{\sqrt{21}} \end{bmatrix}$$

$$\therefore |\hat{a}| = \sqrt{\left(\frac{4}{\sqrt{21}}\right)^2 + \left(\frac{2}{\sqrt{21}}\right)^2 + \left(\frac{1}{\sqrt{21}}\right)^2} = 1$$

$$\hat{b} = \frac{\vec{b}}{|\vec{b}|} = \frac{1}{\sqrt{14}} \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{14} \\ -3/\sqrt{14} \\ 2/\sqrt{14} \end{bmatrix}$$

$$|\hat{b}| = \sqrt{\left(\frac{1}{\sqrt{14}}\right)^2 + \left(\frac{-3}{\sqrt{14}}\right)^2 + \left(\frac{2}{\sqrt{14}}\right)^2} = 1$$

$\therefore$ now $\hat{a}$ and $\hat{b}$ are orthonormal.

$\equiv$ check if S is an orthonormal set.

$$S = \left\{ \frac{1}{\sqrt{5}} (2, 1)^T, \frac{1}{\sqrt{5}} (1, -2)^T \right\}$$

$$= \left\{ \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -2 \end{bmatrix} \right\}$$

$$= \left\{ \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}, \begin{bmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \end{bmatrix} \right\}$$

$\downarrow$ $\downarrow$
 $\vec{u}_1$ $\vec{u}_2$

$$\vec{u}_1 \cdot \vec{u}_2 = \left(\frac{2}{\sqrt{5}} \times \frac{1}{\sqrt{5}} \right) + \left(\frac{1}{\sqrt{5}} \times \frac{-2}{\sqrt{5}} \right) = \frac{2}{5} - \frac{2}{5} = 0$$

$\therefore$ orthogonal

$$|\vec{u}_1| = \sqrt{\left(\frac{2}{\sqrt{5}}\right)^2 + \left(\frac{1}{\sqrt{5}}\right)^2} = 1$$

$$|\vec{u}_2| = \sqrt{\left(\frac{1}{\sqrt{5}}\right)^2 + \left(\frac{-2}{\sqrt{5}}\right)^2} = 1$$

$\therefore S$ is an orthonormal set.

Orthonormal Matrix

The column vectors form an orthonormal set

Q is an orthonormal matrix

properties : $Q^{-1} = Q^T$

$$Q \cdot Q^T = I$$

$$Q^T \cdot Q = I$$

Identity Matrix

3x3

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and

$$\det(I) = 1$$

QR Decomposition - has resuability

$$Ax = b$$

U_1, U_2, U_3 (column's of A matrix)



Gram - Schmidt process →

$$P_1 = U_1$$

P_1, P_2, P_3 (orthogonal)



Normalization

q_1, q_2, q_3 (orthonormal)

column vector



$$A = QR$$

$\downarrow$ $\downarrow$
 $m \times n$ $n \times n$
 $(m > n)$ (upper triangular matrix)
 $\downarrow$
 $m \times n$
(orthonormal matrix)

$$A = QR$$

$$Q^T A = Q^T QR$$

$$Q^T A = IR$$

$$Q^T A = R$$

$$\therefore \boxed{R = Q^T A}$$

$$Ax = b$$

$$A^T Ax = A^T b$$

$$(QR)^T QR x = (QR)^T b$$

$$R^T Q^T QR x = R^T Q^T b$$

$$R^T x = R^T Q^T b$$

$$\therefore \boxed{Rx = Q^T b}$$

can get the value of x using backward substitution.

$$\# \quad x_0 = -3$$

$$x_1 = 0$$

$$x_2 = 6$$

$$f(x_0) = 0$$

$$f(x_1) = 0$$

$$f(x_2) = 2$$

$$A = \begin{pmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{pmatrix}$$

x

$$\begin{pmatrix} a_0 \\ a_1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$$

U_1

U_2

$$P_1 = U_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$P_2 = U_2 - \frac{U_2 \cdot P_1}{P_1 \cdot P_1} P_1 = \begin{bmatrix} -3 \\ 0 \\ 6 \end{bmatrix} - \frac{(-3 \times 1) + (0 \times 1) + (6 \times 1)}{(1 \times 1) + (1 \times 1) + (1 \times 1)} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -3 \\ 0 \\ 6 \end{bmatrix} - 1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -3 \\ 0 \\ 6 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -4 \\ -1 \\ 5 \end{bmatrix}$$

$$\therefore P_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\therefore P_2 = \begin{bmatrix} -4 \\ -1 \\ 5 \end{bmatrix}$$

$$q_1 = \frac{p_1}{|p_1|} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

$$q_2 = \frac{p_2}{|p_2|} = \frac{1}{\sqrt{42}} \begin{bmatrix} -4 \\ -1 \\ 5 \end{bmatrix} = \begin{bmatrix} -4/\sqrt{42} \\ -1/\sqrt{42} \\ 5/\sqrt{42} \end{bmatrix}$$

$$|p_1| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$|p_2| = \sqrt{4^2 + 1^2 + 5^2} = \sqrt{42}$$

$$Q = \begin{bmatrix} 1/\sqrt{3} & -4/\sqrt{42} \\ 1/\sqrt{3} & -1/\sqrt{42} \\ 1/\sqrt{3} & 5/\sqrt{42} \end{bmatrix}$$

$$R = Q^T A$$

$$= \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -4/\sqrt{42} & -1/\sqrt{42} & 5/\sqrt{42} \end{bmatrix} \begin{bmatrix} 1 & 0 & -3 \\ 1 & 0 & 0 \\ 1 & 6 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{bmatrix} \rightarrow \text{upper triangular matrix (2x2)}$$

$$R_x = Q^T b$$

$$\begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{42}} & -\frac{1}{\sqrt{42}} & \frac{5}{\sqrt{42}} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2}{\sqrt{3}} \\ \frac{5/\sqrt{2}}{\sqrt{21}} \end{bmatrix}$$

Backward substitution →

$$\sqrt{42} a_1 = \frac{5/\sqrt{2}}{\sqrt{21}}$$

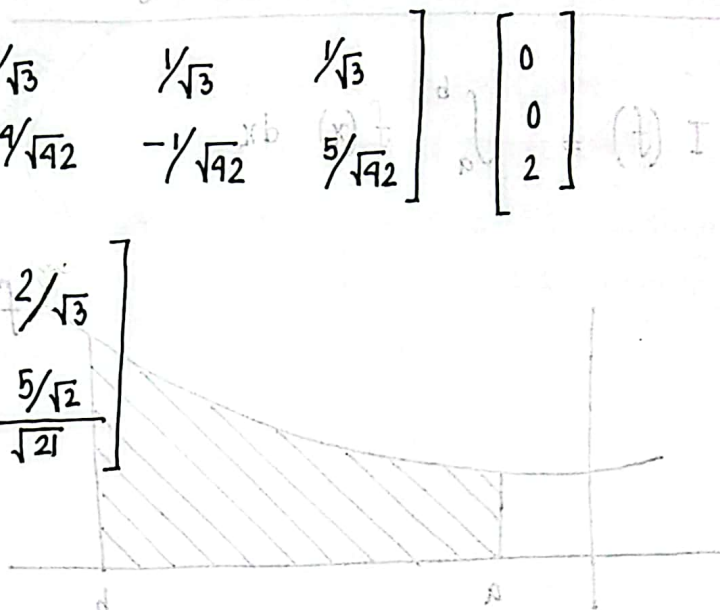
$$\therefore a_1 = 5/21$$

$$\sqrt{3} a_0 + \sqrt{3} a_1 = \frac{2}{\sqrt{3}}$$

$$\therefore a_0 = 3/7$$

$$P_1(x) = a_0 + a_1 x$$

$$P_1(x) = 3/7 + 5/21 x$$



30 April

Assignment 6

Quiz 6

on chapter 6

$$\int_a^b f(x) dx \approx \sum_{k=0}^n w_k f(x_k)$$

$$\int_a^b f(x) dx \approx \sum_{k=0}^n w_k f(x_k)$$

weight factor = w_k

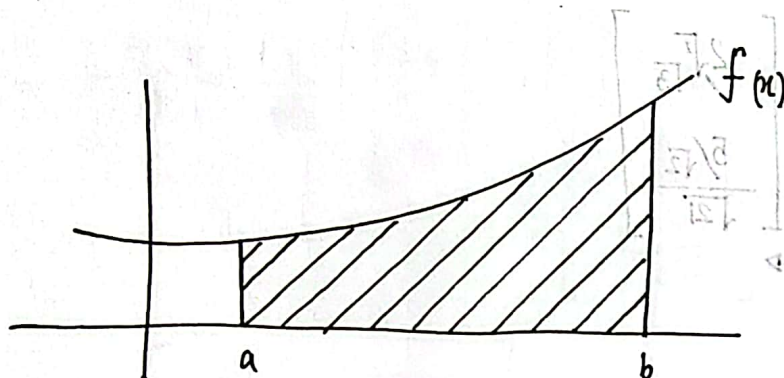
closed (limit)
open

Newton's formula

$$\int_a^b f(x) dx \approx \sum_{k=0}^n w_k f(x_k)$$

chapter 7 : Numerical Integration

$$I(f) = \int_a^b f(x) dx$$



Integration is an infinite sum

$$\text{Actual Integration} = I(f) = \int_a^b f(x) dx$$

$$\text{Numerical Integration} = I_n(f) = \int_a^b P_n(x) dx$$

$$\text{Lagrange Polynomial, } P_n(x) = \sum_{k=0}^n f(x_k) \underbrace{l_k(x)}_{\text{lagrange basis}}$$

$$\therefore I_n(f) = \int_a^b \sum_{k=0}^n f(x_k) l_k(x) dx$$

$$= \sum_{k=0}^n f(x_k) \left[\int_a^b l_k(x) dx \right]$$

$\sigma_k = \text{weighted factor}$

$$\therefore I_n(f) = \sum_{k=0}^n \sigma_k f(x_k) \rightarrow \text{Newton cote's formula} \begin{cases} \text{closed (limit)} \\ \text{open} \end{cases}$$

closed Newton Cote's formula — ① Trapezium/Trapezoidal Rule ($n=1$)

special → ② Simpson's Rule ($n=2$)
[better as degree 2]

→ composite Newton Cote's formula
[accuracy better than ①]

Trapezium/Trapezoidal Rule

degree, $n=1$

number of nodes = 2

↓

$$\left\{ \begin{matrix} x_0 \\ \downarrow a \end{matrix} , \begin{matrix} x_1 \\ \downarrow b \end{matrix} \right\}$$

$$I_n(f) = \int_a^b p_n(x) dx$$

$$I_1(f) = \int_a^b p_1(x) dx$$

$$p_1(x) = l_0(x) f(x_0) + l_1(x) f(x_1)$$

$$I_1(f) = \int_a^b [l_0(x) f(x_0) + l_1(x) f(x_1)] dx$$

$$= f(x_0) \underbrace{\int_a^b l_0(x) dx}_{\sigma_0} + f(x_1) \underbrace{\int_a^b l_1(x) dx}_{\sigma_1}$$

$$= \sigma_0 f(x_0) + \sigma_1 f(x_1)$$

$$\therefore \sigma_0 = \int_a^b l_0(x) dx$$

$$= \int_a^b \frac{x - x_1}{x_0 - x_1} dx$$

closed Newton-Cotes formula

① Trapezoidal Rule ($n=1$) $\int_a^b \frac{x-b}{a-b} dx$

② Simpson's Rule ($n=2$) $\int_a^b \frac{1}{a-b} [(x-b)^2 - (x-a)^2] dx$

composite Newton-Cotes formula

① $\int_a^b \left[\frac{x^2}{2} - bx \right]_a^b$

$= \frac{1}{a-b} \left[\frac{b^2}{2} - b^2 - \frac{a^2}{2} + ab \right]$

$= \frac{b-a}{2}$

degrees, $n=1$

number of nodes = 2

$\downarrow$

$\sigma_1 = \int_a^b l_1(x) dx = \int_a^b \frac{x-x_0}{x_1-x_0} dx = \int_a^b \frac{x-a}{b-a} dx$

$\vdots$

$= \frac{b-a}{2}$

$\therefore I_1(f) = \sigma_0 f(x_0) + \sigma_1 f(x_1)$

$= \frac{b-a}{2} f(a) + \frac{b-a}{2} f(b)$

$I_1 f = \frac{b-a}{2} [f(a) + f(b)] \rightarrow \text{Trapezium Rule}$

$\int_a^b f(x) dx = \underbrace{\int_a^b f(x) dx}_{\sigma_0} + \underbrace{\int_a^b f(x) dx}_{\sigma_1} =$

$\sigma_0 + \sigma_1 =$

$\int_a^b f(x) dx = \sigma_0 + \sigma_1$

$\int_a^b \frac{x-x_1}{x_0-x_1} dx =$

$$\# f(x) = e^x \quad [0, 2]$$

$$I_1(f) = ?$$

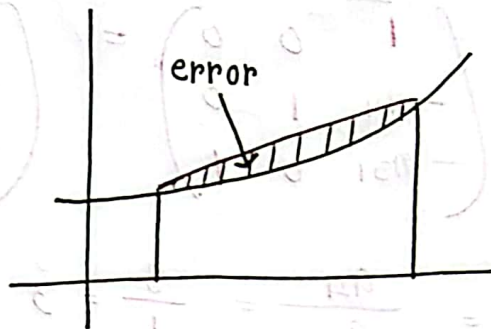
$$\text{Relative error (percentage)} = ?$$

$$\therefore I(f) = \int_0^2 e^x dx$$

$$= [e^x]_0^2$$

$$= e^2 - e^0$$

$$= 6.389$$



$$I_1(f) = \frac{b-a}{2} [f(a) + f(b)]$$

$$= \frac{2-0}{2} [e^0 + e^2]$$

$$= 8.389$$

$$\therefore \text{Relative error} = \left| \frac{\text{actual} - \text{numerical}}{\text{actual}} \right| \times 100\%$$

$$= \left| \frac{I - I_1}{I} \right| \times 100\%$$

$$= \left| \frac{6.389 - 8.389}{6.389} \right| \times 100\%$$

$$= 31.3\% \quad (\text{big error as we are using 1 degree polynomial})$$

closed Newton - Cotes Formula

1. Trapezium Rule

$$n = 1$$

$$a, b$$

$$I_1(f) = \frac{b-a}{2} [f(a) + f(b)]$$

2. Simpson's Rule

$$n = 2$$

$$a, b ; \frac{a+b}{2} = h$$

$$I_2(f) = \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Cauchy's Theorem (upper bound of error)

$$|f(x) - p_n(x)| \leq \left| \frac{f^{(n+1)}(\xi)}{(n+1)!} [(x-x_0)(x-x_1)\dots(x-x_n)] \right|$$

$$|I(f) - I_n(f)| \leq \left| \frac{f^{(n+1)}(\xi)}{(n+1)!} \right| \int_a^b [(x-x_0)(x-x_1)\dots(x-x_n)] dx$$

$$\# n = 1$$

$$f(x) = e^x$$

$$a = 0$$

$$b = 2$$

$$|I - I_1(f)| \leq \left| \frac{f^{(2)}(\xi)}{2!} \right| \int_0^2 |(x-0)(x-2)| dx$$

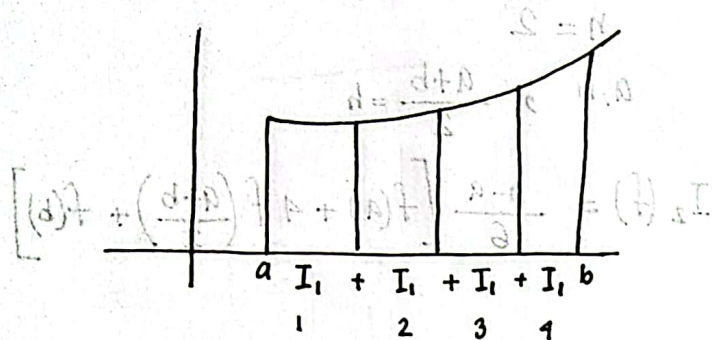
$$\leq \left| \frac{e^{\xi}}{2!} \right| \int_0^2 |x^2 - 2x| dx$$

$$\leq \left| \frac{e^2}{2} \right| \left[\frac{x^3}{3} - \frac{2x^2}{2} \right]_0^2 = \frac{e^2}{3} = 11$$

$$\leq 4.926$$

Composite Newton - Cotes formula

Trapezium but part by part $\rightarrow$ error less



subintervals = $m = 4$

more subintervals, more accurate

$$C_n, m = I_1 + I_2 + I_3 + I_4$$

$n \rightarrow$ degree

$m \rightarrow$ subintervals

$$C_{1,4} = I_1 + I_2 + I_3 + I_4$$

for m intervals, node = $m + 1$

distance between nodes (step size), $h = \frac{b-a}{m}$ [equal distance]

$$C_{1,m}(f) = \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{m-1}) + f(x_m)]$$

$$\# f(x) = e^x$$

$$a = 0$$

$$b = 2$$

$$m = 2$$

$$h = \frac{b-a}{m} = \frac{2-0}{2} = 1$$

$$x_0 = a = 0$$

$$x_1 = a + h = 0 + 1 = 1$$

$$x_2 = b = 2$$

$$C_{1, 2}(f) = \frac{h}{2} [f(x_0) + 2f(x_1) + f(x_2)]$$

$$= \frac{1}{2} [1 + 2e^1 + e^2]$$

$$= 6.91281$$

$$\# f(x) = e^x$$

$$a = 0 \quad h = \frac{b-a}{m} = \frac{2-0}{3} = \frac{2}{3}$$

$$b = 2$$

$$m = 3$$

$$x_0 = a = 0$$

$$x_1 = a + h = \frac{2}{3}$$

$$x_2 = x_1 + h = \frac{2}{3} + \frac{2}{3} = \frac{4}{3}$$

$$x_3 = b = 2$$

$$\therefore C_{1, 3}(f) = \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + f(x_3)]$$

$$= \frac{1}{3} [e^0 + 2e^{\frac{2}{3}} + 2e^{\frac{4}{3}} + e^2]$$

$$= 6.62$$

$$\text{Actual integration, } I(f) = \int_0^2 e^x dx$$

$$= 6.389$$

$C_{1, 3}(f)$ is more close

$\therefore m \uparrow$ accuracy $\uparrow$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 1 \end{bmatrix} \frac{1}{\sqrt{\frac{1}{2} + \frac{1}{2} + 1}} = \frac{\hat{n}}{|\hat{n}|} = \hat{n}$$

$$1 = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + 1^2} = |\hat{n}| \therefore$$